```
"""JobsComing backend application package."""
"""API 路由包。"""
"""平台帳號加密庫（FR-1d / 主程序同步 GAP-2）。

- 寫入：為用戶生成/取回 DEK → 信封加密 → 落庫（密文）
- 讀取：僅返回掩碼後的帳號標識，永不下發密文/明文
- 解密密文只發生在採集/投遞 Worker（P3/P4），通過 get_dek_for_user 取用
"""
from __future__ import annotations

from typing import Annotated

from fastapi import APIRouter, Depends, HTTPException, Request
from sqlalchemy import text
from sqlalchemy.orm import Session

from app.core.audit import log_audit
from app.core.auth import get_current_user
from app.core.security import (
    decrypt_field,
    encrypt_field,
    encrypt_json,
    get_dek_for_user,
)
from app.db import get_db
from app.schemas.accounts import PLATFORMS, AccountCreate, AccountUpdate

router = APIRouter(prefix="/accounts", tags=["accounts"])

CurrentUser = Annotated[dict, Depends(get_current_user)]
DB = Annotated[Session, Depends(get_db)]


def _mask(identifier: str) -> str:
    if len(identifier) <= 4:
        return "****"
    return identifier[:2] + "****" + identifier[-2:]


@router.get("")
def list_accounts(user: CurrentUser, db: DB) -> list[dict]:
    rows = db.execute(
        text(
            "SELECT id, platform, label, login_identifier_enc, status, last_login_at, last_error
"
            "FROM platform_accounts WHERE user_id = :uid ORDER BY created_at"
        ),
        {"uid": user["id"]},
    ).mappings().all()
    dek = get_dek_for_user(db, user["id"])
    result = []
```

```
    for r in rows:
        result.append(
            {
                "id": str(r["id"]),
                "platform": r["platform"],
                "label": r["label"],
                "login_masked": _mask(decrypt_field(dek, r["login_identifier_enc"])),
                "status": r["status"],
                "last_login_at": r["last_login_at"].isoformat() if r["last_login_at"] else None,
                "last_error": r["last_error"],
            }
        )
    return result


@router.post("", status_code=201)
def create_account(user: CurrentUser, db: DB, payload: AccountCreate, request: Request) -> dict:
    if payload.platform not in PLATFORMS:
        raise HTTPException(422, f"platform 必須是 {PLATFORMS} 之一")
    dek = get_dek_for_user(db, user["id"])
    row_id = db.execute(
        text(
            """
            INSERT INTO platform_accounts
              (user_id, platform, label, login_identifier_enc, password_enc, meta_enc)
            VALUES (:uid, :platform, :label, :ident, :pwd, :meta)
            RETURNING id
            """
        ),
        {
            "uid": user["id"],
            "platform": payload.platform,
            "label": payload.label,
            "ident": encrypt_field(dek, payload.login_identifier),
            "pwd": encrypt_field(dek, payload.password),
            "meta": encrypt_json(dek, payload.meta) if payload.meta is not None else None,
        },
    ).mappings().one()
    db.commit()
    log_audit(db, user["id"], "account.create", "platform_accounts", str(row_id["id"]),
              {"platform": payload.platform}, request.client.host if request.client else None)
    return {"id": str(row_id["id"])}


@router.patch("/{account_id}")
def update_account(
    user: CurrentUser, db: DB, account_id: str, payload: AccountUpdate, request: Request
) -> dict:
    exists = db.execute(
        text("SELECT 1 FROM platform_accounts WHERE id = :aid AND user_id = :uid"),
```

```
        {"aid": account_id, "uid": user["id"]},
    ).first()
    if not exists:
        raise HTTPException(404, "帳號不存在")
    dek = get_dek_for_user(db, user["id"])
    fields = payload.model_dump(exclude_none=True)
    sets, params = [], {"aid": account_id, "uid": user["id"]}
    if "label" in fields:
        sets.append("label = :label")
        params["label"] = fields["label"]
    if "login_identifier" in fields:
        sets.append("login_identifier_enc = :ident")
        params["ident"] = encrypt_field(dek, fields["login_identifier"])
    if "password" in fields:
        sets.append("password_enc = :pwd")
        params["pwd"] = encrypt_field(dek, fields["password"])
    if "meta" in fields:
        sets.append("meta_enc = :meta")
        params["meta"] = encrypt_json(dek, fields["meta"]) if fields["meta"] is not None else
None
    if sets:
        sets.append("status = 'pending_verify'")  # 憑據變更後需重新驗證登錄態
        db.execute(
            text(
                f"UPDATE platform_accounts SET {', '.join(sets)}, updated_at = now() "
                "WHERE id = :aid AND user_id = :uid"
            ),
            params,
        )
        db.commit()
        log_audit(db, user["id"], "account.update", "platform_accounts", account_id,
                  {}, request.client.host if request.client else None)
    return {"ok": True}


@router.delete("/{account_id}", status_code=204)
def delete_account(user: CurrentUser, db: DB, account_id: str, request: Request) -> None:
    row = db.execute(
        text("DELETE FROM platform_accounts WHERE id = :aid AND user_id = :uid RETURNING id"),
        {"aid": account_id, "uid": user["id"]},
    ).first()
    if not row:
        raise HTTPException(404, "帳號不存在")
    db.commit()
    log_audit(db, user["id"], "account.delete", "platform_accounts", account_id,
              {}, request.client.host if request.client else None)
"""投遞記錄與觸發端點（FR-3b/3c）。"""
from __future__ import annotations

from typing import Annotated
```

```
from fastapi import APIRouter, Depends, HTTPException
from sqlalchemy import text
from sqlalchemy.orm import Session

from app.core.auth import get_current_user
from app.db import get_db
from app.services.apply_engine import (
    create_applications_from_matches,
    process_queue,
    run_application,
)

router = APIRouter(prefix="/applications", tags=["applications"])

CurrentUser = Annotated[dict, Depends(get_current_user)]
DB = Annotated[Session, Depends(get_db)]


@router.get("")
def list_applications(
    user: CurrentUser, db: DB, status: str | None = None,
    limit: int = 20, offset: int = 0,
) -> dict:
    limit = min(max(limit, 1), 100)
    rows = db.execute(
        text(
            """
            SELECT a.id, a.status, a.status_reason, a.attempts_count, a.next_retry_at,
                   a.submitted_at, a.external_ref, a.created_at,
                   p.company_name, p.title, p.location, p.apply_url, p.source,
                   m.match_score, rv.id AS resume_version_id
            FROM applications a
            JOIN job_postings p ON p.id = a.posting_id
            LEFT JOIN job_matches m ON m.id = a.match_id
            LEFT JOIN resume_versions rv ON rv.id = a.resume_version_id
            WHERE a.user_id = :uid AND (CAST(:st AS text) IS NULL OR CAST(a.status AS text)
= :st)
            ORDER BY a.created_at DESC
            LIMIT :lim OFFSET :off
            """
        ),
        {"uid": user["id"], "st": status, "lim": limit, "off": offset},
    ).mappings().all()
    total = db.execute(
        text(
            """
            SELECT COUNT(*) AS n FROM applications a
            WHERE a.user_id = :uid AND (CAST(:st AS text) IS NULL OR CAST(a.status AS text)
= :st)
            """
        ),
        {"uid": user["id"], "st": status},
```

```
    ).mappings().one()["n"]
    return {
        "items": [
            {
                "id": str(r["id"]),
                "status": r["status"],
                "status_reason": r["status_reason"],
                "attempts_count": r["attempts_count"],
                "next_retry_at": r["next_retry_at"].isoformat() if r["next_retry_at"] else None,
                "submitted_at": r["submitted_at"].isoformat() if r["submitted_at"] else None,
                "external_ref": r["external_ref"],
                "created_at": r["created_at"].isoformat(),
                "company_name": r["company_name"],
                "title": r["title"],
                "location": r["location"],
                "apply_url": r["apply_url"],
                "source": r["source"],
                "match_score": r["match_score"],
                "resume_version_id": str(r["resume_version_id"]) if r["resume_version_id"] else
None,
            }
            for r in rows
        ],
        "total": total,
        "limit": limit,
        "offset": offset,
    }


@router.post("/run", status_code=202)
def run_now(user: CurrentUser, db: DB) -> dict:
    """手動觸發：按當前設置從匹配結果生成投遞隊列並立即處理。"""
    created = create_applications_from_matches(db, user["id"])
    processed = process_queue(db, user_id=user["id"])
    return {"created": created, "processed": processed}


@router.post("/{application_id}/retry", status_code=202)
def retry_application(user: CurrentUser, db: DB, application_id: str) -> dict:
    row = db.execute(
        text("SELECT status FROM applications WHERE id=:id AND user_id=:uid"),
        {"id": application_id, "uid": user["id"]},
    ).mappings().first()
    if not row:
        raise HTTPException(404, "投遞記錄不存在")
    if row["status"] in ("submitted", "in_progress"):
        return {"skipped": row["status"]}
    out = run_application(db, application_id, force=True)
    return out
```

```
@router.post("/{application_id}/abort", status_code=200)
def abort_application(user: CurrentUser, db: DB, application_id: str) -> dict:
    row = db.execute(
        text(
            "UPDATE applications SET status='aborted', status_reason='用戶手動取消',
updated_at=now() "
            "WHERE id=:id AND user_id=:uid AND status NOT IN ('submitted','aborted') RETURNING
id"
        ),
        {"id": application_id, "uid": user["id"]},
    ).first()
    if not row:
        raise HTTPException(404, "投遞記錄不存在或已結束")
    db.commit()
    return {"ok": True}
"""認證代理（關鍵架構）：前端不直連 supabase.co，全部 auth 請求經後端轉發。

原因：部分地區網絡無法直達 *.supabase.co（實測本網絡連接被重置）。
後端部署於美區（Render），可正常訪問 Supabase Auth API。
"""
from __future__ import annotations

import httpx
from fastapi import APIRouter, HTTPException, Request
from pydantic import BaseModel

from app.config import settings

router = APIRouter(prefix="/auth", tags=["auth"])

_SB = settings.supabase_url.rstrip("/")


def _anon_key() -> str:
    return settings.supabase_anon_key


def _anon_headers() -> dict:
    k = _anon_key()
    return {"apikey": k, "Authorization": f"Bearer {k}", "Content-Type": "application/json"}


class Credentials(BaseModel):
    email: str
    password: str


class MagicLinkRequest(BaseModel):
    email: str


class RefreshRequest(BaseModel):
```

```
    refresh_token: str


@router.post("/signup")
def signup(payload: Credentials) -> dict:
    try:
        resp = httpx.post(
            f"{_SB}/auth/v1/signup", headers=_anon_headers(),
            json={"email": payload.email, "password": payload.password}, timeout=20,
        )
    except httpx.HTTPError as exc:
        raise HTTPException(502, f"無法連接 Supabase Auth（服務端網絡問題）: {exc}") from exc
    if resp.status_code >= 400:
        raise HTTPException(resp.status_code, _err(resp))
    data = resp.json()
    if not data.get("session"):
        return {"status": "email_confirmation_required", "user": data.get("user")}
    return {"status": "ok", "session": data["session"], "user": data["user"]}


@router.post("/signin")
def signin(payload: Credentials) -> dict:
    try:
        resp = httpx.post(
            f"{_SB}/auth/v1/token?grant_type=password", headers=_anon_headers(),
            json={"email": payload.email, "password": payload.password}, timeout=20,
        )
    except httpx.HTTPError as exc:
        raise HTTPException(502, f"無法連接 Supabase Auth（服務端網絡問題）: {exc}") from exc
    if resp.status_code >= 400:
        raise HTTPException(resp.status_code, _err(resp))
    return resp.json()


@router.post("/magic-link")
def magic_link(payload: MagicLinkRequest) -> dict:
    try:
        resp = httpx.post(
            f"{_SB}/auth/v1/magiclink", headers=_anon_headers(),
            json={"email": payload.email, "emailRedirectTo": settings.frontend_url}, timeout=20,
        )
    except httpx.HTTPError as exc:
        raise HTTPException(502, f"無法連接 Supabase Auth（服務端網絡問題）: {exc}") from exc
    if resp.status_code >= 400:
        raise HTTPException(resp.status_code, _err(resp))
    return {"status": "sent"}


@router.post("/refresh")
def refresh(payload: RefreshRequest) -> dict:
```

```
    try:
        resp = httpx.post(
            f"{_SB}/auth/v1/token?grant_type=refresh_token", headers=_anon_headers(),
            json={"refresh_token": payload.refresh_token}, timeout=20,
        )
    except httpx.HTTPError as exc:
        raise HTTPException(502, f"無法連接 Supabase Auth（服務端網絡問題）: {exc}") from exc
    if resp.status_code >= 400:
        raise HTTPException(resp.status_code, _err(resp))
    return resp.json()


class AdminCreate(BaseModel):
    email: str
    password: str
    secret: str


@router.post("/admin-create")
def admin_create(payload: AdminCreate) -> dict:
    """維護端點：用 service_role 走 GoTrue 管理 API 正規創建用戶（含郵箱確認）。

  僅供管理員運維使用，secret 由 OPS_SECRET 環境變量保護。
    """
    if not settings.ops_secret or payload.secret != settings.ops_secret:
        raise HTTPException(403, "forbidden")
    try:
        resp = httpx.post(
            f"{_SB}/auth/v1/admin/users",
            headers={
                "apikey": settings.supabase_service_role_key,
                "Authorization": f"Bearer {settings.supabase_service_role_key}",
                "Content-Type": "application/json",
            },
            json={"email": payload.email, "password": payload.password, "email_confirm": True},
            timeout=25,
        )
    except httpx.HTTPError as exc:
        raise HTTPException(502, f"無法連接 Supabase Auth: {exc}") from exc
    try:
        body = resp.json()
    except Exception:  # noqa: BLE001
        body = resp.text[:300]
    return {"status": resp.status_code, "body": body}


def _err(resp: httpx.Response) -> str:
    try:
        return resp.json().get("msg") or resp.json().get("error_description") or resp.text[:300]
    except Exception:  # noqa: BLE001
```

```
        return resp.text[:300]

class ChangePasswordRequest(BaseModel):
    old_password: str
    new_password: str


@router.post("/change-password")
def change_password(user: CurrentUser, db: DB, payload: ChangePasswordRequest, request: Request)
-> dict:
    """修改密碼：先驗證舊密碼，再經 GoTrue 更新。"""
    import httpx as _httpx

    if len(payload.new_password) < 6:
        raise HTTPException(422, "新密碼至少 6 位")
    email = user.get("email")
    # 驗證舊密碼
    try:
        r = _httpx.post(
            f"{settings.supabase_url}/auth/v1/token?grant_type=password",
            headers={"apikey": settings.supabase_anon_key},
            json={"email": email, "password": payload.old_password},
            timeout=20,
        )
    except Exception as exc:  # noqa: BLE001
        raise HTTPException(502, f"驗證服務不可用: {exc}") from exc
    if r.status_code != 200:
        raise HTTPException(401, "舊密碼不正確")
    # 用當前用戶 token 更新密碼
    token = request.headers.get("Authorization", "").removeprefix("Bearer ").strip()
    r = _httpx.put(
        f"{settings.supabase_url}/auth/v1/user",
        headers={"apikey": settings.supabase_anon_key, "Authorization": f"Bearer {token}"},
        json={"password": payload.new_password},
        timeout=20,
    )
    if r.status_code != 200:
        raise HTTPException(502, f"密碼更新失敗: {r.text[:200]}")
    return {"ok": True, "note": "密碼已更新"}
"""關注公司管理 + 公司招聘官網自動發現。"""
from __future__ import annotations

from typing import Annotated

from fastapi import APIRouter, Depends, HTTPException, Request
from pydantic import BaseModel
from sqlalchemy import text
from sqlalchemy.orm import Session

from app.core.audit import log_audit
from app.core.auth import get_current_user
```

```
from app.core.jsonutil import j
from app.db import get_db
from app.schemas.accounts import PLATFORMS, WatchedCompanyCreate, WatchedCompanyUpdate

router = APIRouter(prefix="/companies", tags=["companies"])

CurrentUser = Annotated[dict, Depends(get_current_user)]
DB = Annotated[Session, Depends(get_db)]


@router.get("")
def list_companies(user: CurrentUser, db: DB) -> list[dict]:
    rows = db.execute(
        text(
            """
            SELECT DISTINCT ON (COALESCE(careers_url, company_name)) id, company_name, aliases,
                   careers_url, platform, notes, is_active, created_at
            FROM watched_companies WHERE user_id = :uid
            ORDER BY COALESCE(careers_url, company_name), created_at
            """
        ),
        {"uid": user["id"]},
    ).mappings().all()
    return [
        {
            "id": str(r["id"]),
            "company_name": r["company_name"],
            "aliases": r["aliases"] or [],
            "careers_url": r["careers_url"],
            "platform": r["platform"],
            "notes": r["notes"],
            "is_active": r["is_active"],
            "created_at": r["created_at"].isoformat(),
        }
        for r in rows
    ]


@router.post("", status_code=201)
def create_company(
    user: CurrentUser, db: DB, payload: WatchedCompanyCreate, request: Request
) -> dict:
    if payload.platform not in PLATFORMS:
        raise HTTPException(422, f"platform 必须是{PLATFORMS} 之一")
    row = db.execute(
        text(
            """
            INSERT INTO watched_companies
              (user_id, company_name, aliases, careers_url, platform, notes)
            VALUES (:uid, :name, CAST(:aliases AS jsonb), :url, :platform, :notes)
```

```
            RETURNING id
            """
        ),
        {
            "uid": user["id"],
            "name": payload.company_name,
            "aliases": j(payload.aliases or []),
            "url": payload.careers_url,
            "platform": payload.platform,
            "notes": payload.notes,
        },
    ).mappings().one()
    db.commit()
    log_audit(db, user["id"], "company.create", "watched_companies", str(row["id"]),
              {"name": payload.company_name}, request.client.host if request.client else None)
    return {"id": str(row["id"])}


@router.patch("/{company_id}")
def update_company(
    user: CurrentUser, db: DB, company_id: str, payload: WatchedCompanyUpdate
) -> dict:
    fields = payload.model_dump(exclude_none=True)
    if "aliases" in fields:
        fields["aliases"] = j(fields["aliases"] or [])
    sets = ", ".join(f"{k} = CAST(:{k} AS jsonb)" if k == "aliases" else f"{k} = :{k}"
                     for k in fields)
    result = db.execute(
        text(
            f"UPDATE watched_companies SET {sets}, updated_at = now() "
            "WHERE id = :cid AND user_id = :uid RETURNING id"
        ) if sets else text("SELECT id FROM watched_companies WHERE id = :cid AND user_id
= :uid"),
        {"cid": company_id, "uid": user["id"], **fields},
    )
    db.commit()
    if not result.first():
        raise HTTPException(404, "公司不存在")
    return {"ok": True}


@router.delete("/{company_id}", status_code=204)
def delete_company(user: CurrentUser, db: DB, company_id: str) -> None:
    row = db.execute(
        text("DELETE FROM watched_companies WHERE id = :cid AND user_id = :uid RETURNING id"),
        {"cid": company_id, "uid": user["id"]},
    ).first()
    if not row:
        raise HTTPException(404, "公司不存在")
    db.commit()
```

```
class DiscoverRequest(BaseModel):
    name: str


@router.post("/discover")
def discover_company(user: CurrentUser, payload: DiscoverRequest) -> dict:
    """按公司名自動發現招聘官網：LLM 給候選 + 實網探測驗證。"""
    import re

    import httpx as _httpx

    from app.core.llm import chat_json

    name = payload.name.strip()[:100]
    out = chat_json(
        "你熟悉各公司招聘官網。給出公司招聘官網地址候選。輸出 JSON: "
        '{"candidates": ["url1", "url2", ...], "careers_hint": "一句話說明"}',
        f"公司名：{name}",
        max_tokens=500,
    )
    verified = []
    for url in (out.get("candidates") or [])[:6]:
        if not url.startswith("http"):
            continue
        try:
            r = _httpx.get(url, timeout=12, follow_redirects=True,
                           headers={"User-Agent": "Mozilla/5.0"})
            if r.status_code == 200:
                m = re.search(r"<title>([^<]*)</title>", r.text, re.I)
                verified.append({"url": str(r.url), "status": 200,
                                 "title": (m.group(1).strip()[:80] if m else None)})
            else:
                verified.append({"url": url, "status": r.status_code, "title": None})
        except Exception as exc:  # noqa: BLE001
            verified.append({"url": url, "status": "unreachable", "title": str(exc)[:80]})
    return {"name": name, "hint": out.get("careers_hint"), "candidates": verified}
"""個人信息元信息表（投遞表單互補機制）。

- 用戶維護一份「表單問答庫」：key（字段名/問題）→ value
- 投遞遇到表單必填項時：精確 key 命中直接填；相似度 ≥0.5 自動填（記入 attempt trace）
- 人工接管回填的答案自動存庫（source=manual_answer），下次免問
- 與簡歷解析結果互補：簡歷有 → 畫像層；簡歷沒有 → 元信息表 → 問用戶

保護方式：
- 敏感字段由服務端用該用戶的數據密鑰做 AES-256-GCM 字段級加密後落庫（sensitive_enc）
- 列表接口只返回掩碼值；明文僅在「查看」與「填寫方案」按需解密，不批量下發
"""
from __future__ import annotations
```

```
import difflib
import re
from typing import Annotated

from fastapi import APIRouter, Depends, HTTPException
from pydantic import BaseModel
from sqlalchemy import text
from sqlalchemy.orm import Session

from app.core.auth import get_current_user
from app.core.security import decrypt_field, encrypt_field, get_dek_for_user
from app.db import get_db

router = APIRouter(prefix="/me/info-fields", tags=["info-fields"])

CurrentUser = Annotated[dict, Depends(get_current_user)]
DB = Annotated[Session, Depends(get_db)]


class InfoFieldUpsert(BaseModel):
    key: str
    value: str


class InfoFieldItem(BaseModel):
    key: str
    value: str


class InfoFieldsBatch(BaseModel):
    fields: list[InfoFieldItem]


SENSITIVE_KEY_RE = re.compile(
    r"證件|证件|身份證|身份证|護照|护照|家庭|親屬|亲属|緊急聯繫|紧急联系|"
    r"證件號|证件号|地址|邮编|郵編|電話|电话|手機|手机|姓名拼音|父|母"
)


def _mask_value(v: str) -> str:
    """敏感字段服務器端只存掩碼（列表顯示用），明文不落庫。"""
    if len(v) <= 4:
        return "****"
    return v[:2] + "*" * min(len(v) - 4, 8) + v[-2:]


def _plaintext(db: Session, user_id: str, value: str, enc: str | None) -> str:
    """有密文則解密，否則返回存的明文（非敏感字段）。解密失敗回退掩碼值。"""
    if not enc:
        return value
```

```
    try:
        return decrypt_field(get_dek_for_user(db, user_id), enc)
    except Exception:  # noqa: BLE001
        return value


@router.get("")
def list_fields(user: CurrentUser, db: DB) -> list[dict]:
    rows = db.execute(
        text(
            "SELECT key, value, sensitive_enc, source, updated_at FROM user_info_fields "
            "WHERE user_id=:uid ORDER BY updated_at DESC LIMIT 200"
        ),
        {"uid": user["id"]},
    ).mappings().all()
    return [
        {"key": r["key"], "value": r["value"], "source": r["source"],
         "updated_at": r["updated_at"].isoformat(),
         "sensitive": bool(SENSITIVE_KEY_RE.search(r["key"]))}
        for r in rows
    ]


@router.get("/{key}/reveal")
def reveal_field(user: CurrentUser, db: DB, key: str) -> dict:
    """按需查看單個字段明文（僅本人、僅單條），用於「顯示」按鈕。"""
    row = db.execute(
        text("SELECT value, sensitive_enc FROM user_info_fields WHERE user_id=:uid AND key=:k"),
        {"uid": user["id"], "k": key.strip()[:100]},
    ).mappings().first()
    if not row:
        raise HTTPException(404, "字段不存在")
    return {"key": key, "value": _plaintext(db, user["id"], row["value"], row["sensitive_enc"])}


@router.put("")
def upsert_field(user: CurrentUser, db: DB, payload: InfoFieldUpsert) -> dict:
    key = payload.key.strip()[:100]
    value = payload.value.strip()[:2000]
    sensitive = bool(SENSITIVE_KEY_RE.search(key))
    if sensitive:
        enc = encrypt_field(get_dek_for_user(db, user["id"]), value)
        db.execute(
            text(
                """
                INSERT INTO user_info_fields (user_id, key, value, sensitive_enc, source)
                VALUES (:uid, :k, :v, :enc, 'user')
                ON CONFLICT (user_id, key) DO UPDATE
                  SET value=EXCLUDED.value, sensitive_enc=EXCLUDED.sensitive_enc,
                      source='user', updated_at=now()
```

```
                """
            ),
            {"uid": user["id"], "k": key, "v": _mask_value(value), "enc": enc[:4000]},
        )
    else:
        db.execute(
            text(
                """
                INSERT INTO user_info_fields (user_id, key, value, source)
                VALUES (:uid, :k, :v, 'user')
                ON CONFLICT (user_id, key) DO UPDATE
                  SET value=EXCLUDED.value, sensitive_enc=NULL,
                      source='user', updated_at=now()
                """
            ),
            {"uid": user["id"], "k": key, "v": value},
        )
    db.commit()
    return {"ok": True, "sensitive": sensitive}


@router.put("/batch")
def upsert_fields_batch(user: CurrentUser, db: DB, payload: InfoFieldsBatch) -> dict:
    """批量寫入（投遞回填 / 資料匯入）：敏感字段由服務端加密後落庫。"""
    saved = 0
    for f in payload.fields:
        key = f.key.strip()[:100]
        if not key:
            continue
        value = f.value.strip()[:2000]
        sensitive = bool(SENSITIVE_KEY_RE.search(key))
        if sensitive:
            enc = encrypt_field(get_dek_for_user(db, user["id"]), value)[:4000]
            stored, mask = _mask_value(value), enc
        else:
            stored, mask = value, None
        db.execute(
            text(
                """
                INSERT INTO user_info_fields (user_id, key, value, sensitive_enc, source)
                VALUES (:uid, :k, :v, :enc, 'website')
                ON CONFLICT (user_id, key) DO UPDATE
                  SET value=EXCLUDED.value,
                      sensitive_enc=COALESCE(EXCLUDED.sensitive_enc,
user_info_fields.sensitive_enc),
                      source='website', updated_at=now()
                """
            ),
            {"uid": user["id"], "k": key, "v": stored, "enc": mask},
        )
        saved += 1
```

```
    db.commit()
    return {"ok": True, "saved": saved}


@router.delete("/{key}")
def delete_field(user: CurrentUser, db: DB, key: str) -> dict:
    db.execute(
        text("DELETE FROM user_info_fields WHERE user_id=:uid AND key=:k"),
        {"uid": user["id"], "k": key},
    )
    db.commit()
    return {"ok": True}


# ── 相似度解析（供投遞引擎調用）──────────────────────────────────────

def _tokens(s: str) -> set[str]:
    s = re.sub(r"[^\w\u4e00-\u9fff]+", " ", s.lower())
    toks = {t for t in s.split() if len(t) > 1}
    # 中文按單字+雙字組拆
    for w in re.findall(r"[\u4e00-\u9fff]+", s):
        toks.update(w)
        toks.update(w[i:i + 2] for i in range(len(w) - 1))
    return toks


def similarity(a: str, b: str) -> float:
    """字段名/問題的相似度：序列比對 + token 重疊取較大者。"""
    if not a or not b:
        return 0.0
    if a.strip().lower() == b.strip().lower():
        return 1.0
    seq = difflib.SequenceMatcher(None, a.lower(), b.lower()).ratio()
    ta, tb = _tokens(a), _tokens(b)
    if not ta or not tb:
        return seq
    overlap = len(ta & tb) / min(len(ta), len(tb))
    return max(seq, overlap)


def resolve_field(db: Session, user_id: str, field_key: str, field_label: str,
                  threshold: float = 0.5) -> tuple[str | None, str | None, float]:
    """為表單字段找答案：精確命中 → 相似度 ≥ 閾值 → 無。

  返回 (value, 來源 key, 相似度)；value=None 表示需要詢問用戶。
  敏感字段在此按需解密，保證自動填報拿到的是真實內容而非掩碼。
    """
    rows = db.execute(
        text("SELECT key, value, sensitive_enc, source FROM user_info_fields WHERE
user_id=:uid"),
        {"uid": user_id},
```

```
    ).mappings().all()
    if not rows:
        return None, None, 0.0

    def plain(r) -> str:
        return _plaintext(db, user_id, r["value"], r["sensitive_enc"])

    # 精確命中（key 或 label 一致）
    for r in rows:
        if r["key"].strip().lower() in (field_key.lower(), field_label.lower()):
            return plain(r), r["key"], 1.0
    # 相似度匹配
    best = max(rows, key=lambda r: max(
        similarity(r["key"], field_key), similarity(r["key"], field_label)))
    score = max(similarity(best["key"], field_key), similarity(best["key"], field_label))
    if score >= threshold:
        return plain(best), best["key"], round(score, 2)
    return None, None, round(score, 2)


def save_answer(db: Session, user_id: str, key: str, value: str) -> None:
    """人工回填答案自動入庫（投遞引擎調用）；敏感字段同樣加密後落庫。"""
    key = key.strip()[:100]
    value = value.strip()[:2000]
    if SENSITIVE_KEY_RE.search(key):
        enc = encrypt_field(get_dek_for_user(db, user_id), value)
        stored = _mask_value(value)
    else:
        enc, stored = None, value
    db.execute(
        text(
            """
            INSERT INTO user_info_fields (user_id, key, value, sensitive_enc, source)
            VALUES (:uid, :k, :v, :enc, 'manual_answer')
            ON CONFLICT (user_id, key) DO UPDATE
              SET value=EXCLUDED.value,
                  sensitive_enc=COALESCE(EXCLUDED.sensitive_enc,
user_info_fields.sensitive_enc),
                  source='manual_answer', updated_at=now()
            """
        ),
        {"uid": user_id, "k": key, "v": stored, "enc": enc},
    )
"""崗位瀏覽與手動採集端點（FR-3a）。"""
from __future__ import annotations

import logging
from typing import Annotated

from fastapi import APIRouter, BackgroundTasks, Depends, HTTPException
from pydantic import BaseModel
```

```
from sqlalchemy import text
from sqlalchemy.orm import Session

from app.core.auth import get_current_user
from app.core.jsonutil import j
from app.db import get_db

logger = logging.getLogger(__name__)

router = APIRouter(prefix="/jobs", tags=["jobs"])

CurrentUser = Annotated[dict, Depends(get_current_user)]
DB = Annotated[Session, Depends(get_db)]


class CollectRequest(BaseModel):
    company: str | None = None


@router.get("")
def list_jobs(
    user: CurrentUser, db: DB, q: str | None = None, company: str | None = None,
    campus: str | None = None, org: str | None = None,
    limit: int = 20, offset: int = 0,
) -> dict:
    limit = min(max(limit, 1), 200)
    rows = db.execute(
        text(
            """
            SELECT p.id, p.company_name, p.title, p.location, p.department, p.source,
                   p.apply_url, p.first_seen_at, p.posted_at, p.is_campus, p.org_level,
                   m.match_score, m.priority_score, m.status AS match_status, m.id AS match_id
            FROM job_postings p
            LEFT JOIN job_matches m ON m.posting_id = p.id AND m.user_id = :uid
            WHERE p.is_active = true
              AND (CAST(:q AS text) IS NULL OR p.title ILIKE '%'||:q||'%' OR p.description_text
ILIKE '%'||:q||'%')
              AND (CAST(:co AS text) IS NULL OR p.company_name ILIKE '%'||:co||'%')
              AND (CAST(:ca AS text) IS NULL OR (CAST(:ca AS text)='campus' AND p.is_campus)
                   OR (CAST(:ca AS text)='society' AND NOT p.is_campus))
              AND (CAST(:og AS text) IS NULL OR p.org_level = :og)
            ORDER BY p.first_seen_at DESC
            LIMIT :lim OFFSET :off
            """
        ),
        {"uid": user["id"], "q": q, "co": company, "ca": campus, "og": org,
         "lim": limit, "off": offset},
    ).mappings().all()
    total = db.execute(
        text(
            """
```

```
            SELECT COUNT(*) AS n FROM job_postings p
            WHERE p.is_active = true
              AND (CAST(:q AS text) IS NULL OR p.title ILIKE '%'||:q||'%' OR p.description_text
ILIKE '%'||:q||'%')
              AND (CAST(:co AS text) IS NULL OR p.company_name ILIKE '%'||:co||'%')
              AND (CAST(:ca AS text) IS NULL OR (CAST(:ca AS text)='campus' AND p.is_campus)
                   OR (CAST(:ca AS text)='society' AND NOT p.is_campus))
              AND (CAST(:og AS text) IS NULL OR p.org_level = :og)
            """
        ),
        {"q": q, "co": company, "ca": campus, "og": org},
    ).mappings().one()["n"]
    return {
        "items": [
            {
                "id": str(r["id"]),
                "company_name": r["company_name"],
                "title": r["title"],
                "location": r["location"],
                "department": r["department"],
                "source": r["source"],
                "apply_url": r["apply_url"],
                "first_seen_at": r["first_seen_at"].isoformat(),
                "posted_at": r["posted_at"].isoformat() if r["posted_at"] else None,
                "match_score": r["match_score"],
                "priority_score": r["priority_score"],
                "is_campus": r["is_campus"],
                "org_level": r["org_level"],
                "match_status": r["match_status"],
                "match_id": str(r["match_id"]) if r["match_id"] else None,
            }
            for r in rows
        ],
        "limit": limit,
        "offset": offset,
        "total": total,
    }


@router.get("/collect-status")
def collect_status(user: CurrentUser, db: DB) -> dict:
    """採集進度：前台輪詢此端點，running=false 且 finished_at 有值即完成。"""
    row = db.execute(
        text("SELECT collect_state FROM user_settings WHERE user_id=:u"),
        {"u": user["id"]},
    ).mappings().first()
    return dict(row["collect_state"] or {}) if row and row["collect_state"] else {"running":
False}


@router.get("/{posting_id}")
def get_job(user: CurrentUser, db: DB, posting_id: str) -> dict:
```

```
    row = db.execute(
        text(
            """
            SELECT p.*, m.match_score, m.reasons, m.hard_failures, m.status AS match_status
            FROM job_postings p
            LEFT JOIN job_matches m ON m.posting_id = p.id AND m.user_id = :uid
            WHERE p.id = :pid
            """
        ),
        {"uid": user["id"], "pid": posting_id},
    ).mappings().first()
    if not row:
        raise HTTPException(404, "崗位不存在")
    return {
        "id": str(row["id"]),
        "company_name": row["company_name"],
        "title": row["title"],
        "location": row["location"],
        "department": row["department"],
        "source": row["source"],
        "apply_url": row["apply_url"],
        "description_text": row["description_text"],
        "first_seen_at": row["first_seen_at"].isoformat(),
        "match_score": row["match_score"],
        "match_status": row["match_status"],
        "reasons": row["reasons"] or [],
        "hard_failures": row["hard_failures"] or [],
    }


@router.post("/collect/run", status_code=202)
def run_collect(user: CurrentUser, db: DB, payload: CollectRequest, background_tasks:
BackgroundTasks) -> dict:
    """異步觸發：全量或按公司採集 + 匹配（後台執行）。"""
    from app.collectors.pipeline import collect_for_user

    uid = user["id"]
    company = (payload.company or "").strip()

    def _run() -> None:
        import datetime as _dt

        from app.db import SessionLocal

        s = SessionLocal()
        s.execute(
            text(
                "INSERT INTO user_settings (user_id, collect_state) VALUES (:u, CAST(:st AS
jsonb)) "
                "ON CONFLICT (user_id) DO UPDATE SET collect_state=CAST(:st AS jsonb),
updated_at=now()"
            ),
            {"u": uid, "st": j({"running": True, "finished_at": None, "report": None})},
```

```
        )
        s.commit()
        try:
            report = collect_for_user(s, uid, company_name=company or None)
            s.execute(
                text(
                    "UPDATE user_settings SET collect_state=CAST(:st AS jsonb), updated_at=now()
"
                    "WHERE user_id=:u"
                ),
                {"u": uid, "st": j({
                    "running": False,
                    "finished_at": _dt.datetime.now(_dt.timezone.utc).isoformat(),
                    "report": {k: report.get(k) for k in ("new", "updated", "matched", "errors",
"collected")},
                })},
            )
            s.commit()
        except Exception:  # noqa: BLE001
            logger.exception("後台採集失敗 user=%s", uid)
            s.execute(
                text(
                    "UPDATE user_settings SET collect_state=CAST(:st AS jsonb), updated_at=now()
"
                    "WHERE user_id=:u"
                ),
                {"u": uid, "st": j({"running": False, "finished_at": None, "report": {"errors":
1}})},
            )
            s.commit()
        finally:
            s.close()

    background_tasks.add_task(_run)
    return {"started": True, "note": ("僅採集該公司" if company else "全量採集") + "，後台執行中"}


@router.get("/collect-status")
def collect_status(user: CurrentUser, db: DB) -> dict:
    """採集進度：前台輪詢此端點，running=false 且 finished_at 有值即完成。"""
    row = db.execute(
        text("SELECT collect_state FROM user_settings WHERE user_id=:u"),
        {"u": user["id"]},
    ).mappings().first()
    return dict(row["collect_state"] or {}) if row and row["collect_state"] else {"running":
False}
"""崗位匹配端點（FR-1f）。採集流水線（P3）也復用 score_posting。"""
from __future__ import annotations

import logging
from typing import Annotated

from fastapi import APIRouter, BackgroundTasks, Depends, HTTPException
from pydantic import BaseModel
from sqlalchemy import text
```

```
from sqlalchemy.orm import Session

from app.core.auth import get_current_user
from app.core.jsonutil import loads_json
from app.db import get_db
from app.services.matching import score_posting

logger = logging.getLogger(__name__)

router = APIRouter(prefix="/matches", tags=["matches"])

CurrentUser = Annotated[dict, Depends(get_current_user)]
DB = Annotated[Session, Depends(get_db)]

REMATCH_LIMIT = 200  # 每輪評分上限（控制 LLM 成本）


def rematch_all_for_user(db: Session, user_id: str, limit: int = 30) -> dict:
    """用當前畫像對用戶匹配做一輪評分（補缺失優先 + 重算最舊的），返回進度。

  準確性 100% 策略：分批多輪直到 remaining=0；
  每輪優先補評「完全沒有 match 行」的崗位，再重算最舊的已有 match。
    """
    total = 0
    # 1) 補評缺失 match 的崗位（採集時漏評/新公司）
    missing = db.execute(
        text(
            """
            SELECT p.id FROM job_postings p
            LEFT JOIN job_matches m ON m.posting_id = p.id AND m.user_id = :u
            WHERE m.id IS NULL AND p.is_active = true
            ORDER BY p.first_seen_at DESC LIMIT :n
            """
        ),
        {"u": user_id, "n": limit},
    ).mappings().all()
    for r in missing:
        try:
            score_posting(db, user_id, str(r["id"]))
            total += 1
        except Exception:  # noqa: BLE001
            logger.exception("補評失敗 user=%s posting=%s", user_id, r["id"])
    # 2) 重算最舊的已有 match
    if len(missing) < limit:
        stale = db.execute(
            text(
                "SELECT posting_id FROM job_matches WHERE user_id=:u "
                "ORDER BY updated_at ASC LIMIT :n"
            ),
            {"u": user_id, "n": limit - len(missing)},
```

```
        ).mappings().all()
        for r in stale:
            try:
                score_posting(db, user_id, str(r["posting_id"]))
                total += 1
            except Exception:  # noqa: BLE001
                logger.exception("重新匹配失敗 user=%s posting=%s", user_id, r["posting_id"])
    # 3) 剩餘量（供進度查詢/多輪調度）
    remaining = db.execute(
        text(
            """
            SELECT (SELECT COUNT(*) FROM job_postings p
                    LEFT JOIN job_matches m ON m.posting_id = p.id AND m.user_id = :u
                    WHERE m.id IS NULL AND p.is_active = true) AS miss,
                   (SELECT COUNT(*) FROM job_matches WHERE user_id = :u) AS has
            """
        ),
        {"u": user_id},
    ).mappings().one()
    return {"scored": total, "remaining_missing": int(remaining["miss"]),
            "total_matches": int(remaining["has"])}


@router.post("/{posting_id}/score", status_code=201)
def score(user: CurrentUser, db: DB, posting_id: str) -> dict:
    try:
        return score_posting(db, user["id"], posting_id)
    except ValueError as exc:
        raise HTTPException(400, str(exc)) from exc


@router.get("/companies")
def list_companies(user: CurrentUser, db: DB) -> list[dict]:
    """匹配頁公司篩選列表（公司 + 匹配數 + 最高分；含 hard_failed 的公司）。"""
    rows = db.execute(
        text(
            """
            SELECT p.company_name, COUNT(*) AS n, MAX(m.match_score) AS top
            FROM job_matches m JOIN job_postings p ON p.id = m.posting_id
            WHERE m.user_id = :uid
            GROUP BY p.company_name ORDER BY top DESC NULLS LAST, n DESC LIMIT 100
            """
        ),
        {"uid": user["id"]},
    ).mappings().all()
    return [
        {"company_name": r["company_name"], "count": r["n"], "top_score": r["top"]}
        for r in rows
    ]
```

```
def rematch_company(db: Session, user_id: str, company: str, batch: int = 50) -> dict:
    """按公司重新匹配：分批循環直到該公司全部崗位重算。"""
    total = 0
    for _ in range(20):  # 每輪 50 個，最多 1000 個/公司
        rows = db.execute(
            text(
                """
                SELECT p.id FROM job_postings p
                WHERE p.is_active = true AND p.company_name ILIKE '%'||:co||'%'
                ORDER BY p.first_seen_at DESC LIMIT :n OFFSET :off
                """
            ),
            {"co": company, "n": batch, "off": total},
        ).mappings().all()
        if not rows:
            break
        for r in rows:
            try:
                score_posting(db, user_id, str(r["id"]))
                total += 1
            except Exception:  # noqa: BLE001
                logger.exception("按公司匹配失敗 user=%s posting=%s", user_id, r["id"])
    return {"scored": total, "company": company, "remaining_missing": 0, "total_matches": total}


def rematch_full(db: Session, user_id: str, max_rounds: int = 30, batch: int = 30) -> dict:
    """分批多輪直到全量補評完成。返回最終進度。"""
    res = {"scored": 0, "remaining_missing": -1, "total_matches": 0}
    for _round in range(max_rounds):
        res = rematch_all_for_user(db, user_id, limit=batch)
        if res["remaining_missing"] == 0:
            break
    return res


class RematchRequest(BaseModel):
    company: str | None = None


@router.post("/rematch", status_code=202)
def rematch(user: CurrentUser, db: DB, payload: RematchRequest, background_tasks:
BackgroundTasks) -> dict:
    """重新匹配：全量或按公司（補評缺失 + 重算），後台執行。"""
    uid = user["id"]
    company = (payload.company or "").strip()

    def _run() -> None:
        from app.db import SessionLocal

        s = SessionLocal()
```

```
        try:
            if company:
                res = rematch_company(s, uid, company)
            else:
                res = rematch_full(s, uid)
            logger.info("重新匹配完成 user=%s %s", uid, res)
        except Exception:  # noqa: BLE001
            logger.exception("重新匹配失敗 user=%s", uid)
        finally:
            s.close()

    background_tasks.add_task(_run)
    return {"started": True, "note": ("僅重新匹配該公司" if company else "全量重新匹配") + "，後台執行
中"}


@router.get("/rematch-status")
def rematch_status(user: CurrentUser, db: DB) -> dict:
    """重新匹配進度：未評分崗位數 + 已有匹配數。"""
    row = db.execute(
        text(
            """
            SELECT (SELECT COUNT(*) FROM job_postings p
                    LEFT JOIN job_matches m ON m.posting_id = p.id AND m.user_id = :u
                    WHERE m.id IS NULL AND p.is_active = true) AS miss,
                   (SELECT COUNT(*) FROM job_matches WHERE user_id = :u) AS has
            """
        ),
        {"u": user["id"]},
    ).mappings().one()
    return {"remaining_missing": int(row["miss"]), "total_matches": int(row["has"])}


@router.get("")
def list_matches(
    user: CurrentUser, db: DB, status: str | None = None,
    company: str | None = None,
    limit: int = 20, offset: int = 0,
    campus: str | None = None, org: str | None = None,
) -> dict:
    limit = min(max(limit, 1), 100)
    rows = db.execute(
        text(
            """
            SELECT m.id, m.posting_id, m.match_score, m.priority_score, m.hard_failures,
                   m.reasons, m.score_breakdown, m.direction_scores, m.status,
                   p.company_name, p.title, p.location, p.department, p.is_campus,
                   p.org_level, p.apply_url, p.first_seen_at, p.raw->'directions' AS directions
            FROM job_matches m
            JOIN job_postings p ON p.id = m.posting_id
            WHERE m.user_id = :uid AND (CAST(:st AS text) IS NULL OR CAST(m.status AS text)
= :st)
```

```
              AND (CAST(:co AS text) IS NULL OR p.company_name ILIKE '%'||:co||'%')
              AND (CAST(:ca AS text) IS NULL OR (CAST(:ca AS text)='campus' AND p.is_campus)
                   OR (CAST(:ca AS text)='society' AND NOT p.is_campus))
              AND (CAST(:og AS text) IS NULL OR p.org_level = :og)
            ORDER BY m.priority_score DESC NULLS LAST, m.match_score DESC NULLS LAST,
                     p.first_seen_at DESC
            LIMIT :lim OFFSET :off
            """
        ),
        {"uid": user["id"], "st": status, "co": company, "ca": campus, "og": org,
         "lim": limit, "off": offset},
    ).mappings().all()
    total = db.execute(
        text(
            """
            SELECT COUNT(*) AS n FROM job_matches m
            JOIN job_postings p ON p.id = m.posting_id
            WHERE m.user_id = :uid AND (CAST(:st AS text) IS NULL OR CAST(m.status AS text)
= :st)
              AND (CAST(:co AS text) IS NULL OR p.company_name ILIKE '%'||:co||'%')
              AND (CAST(:ca AS text) IS NULL OR (CAST(:ca AS text)='campus' AND p.is_campus)
                   OR (CAST(:ca AS text)='society' AND NOT p.is_campus))
              AND (CAST(:og AS text) IS NULL OR p.org_level = :og)
            """
        ),
        {"uid": user["id"], "st": status, "co": company, "ca": campus, "og": org},
    ).mappings().one()["n"]
    return {
        "items": [
            {
                "id": str(r["id"]),
                "posting_id": str(r["posting_id"]),
                "match_score": r["match_score"],
                "priority_score": r["priority_score"],
                "is_campus": r["is_campus"],
                "org_level": r["org_level"],
                "department": r["department"],
                "directions": loads_json(r["directions"])
                if isinstance(r["directions"], str) else (r["directions"] or []),
                "hard_failures": r["hard_failures"] or [],
                "score_breakdown": r["score_breakdown"] or {},
                "direction_scores": loads_json(r["direction_scores"])
                if isinstance(r["direction_scores"], str)
                else (r["direction_scores"] or []),
                "reasons": r["reasons"] or [],
                "status": r["status"],
                "company_name": r["company_name"],
                "title": r["title"],
                "location": r["location"],
                "apply_url": r["apply_url"],
                "first_seen_at": r["first_seen_at"].isoformat(),
```

```
            }
            for r in rows
        ],
        "total": total,
        "limit": limit,
        "offset": offset,
    }
"""人工接管任務端點（FR-3c：直接框 → 回填 → 繼續投遞）。"""
from __future__ import annotations

from typing import Annotated

from fastapi import APIRouter, Depends, HTTPException
from pydantic import BaseModel
from sqlalchemy import text
from sqlalchemy.orm import Session

from app.core.auth import get_current_user
from app.db import get_db
from app.services.apply_engine import answer_and_resume

router = APIRouter(prefix="/manual-tasks", tags=["manual-tasks"])

CurrentUser = Annotated[dict, Depends(get_current_user)]
DB = Annotated[Session, Depends(get_db)]


class TaskAnswer(BaseModel):
    answer: dict | None = None


@router.get("/{task_id}/fill-plan")
def task_fill_plan(user: CurrentUser, db: DB, task_id: str) -> dict:
    """填寫方案：為人工投遞生成按當前資料整理好的答案，可一鍵複製到官網表單。"""
    from app.services.apply_engine import _load_application
    from app.services.form_mapping import build_candidate_payload

    task = db.execute(
        text("SELECT application_id FROM manual_tasks WHERE id=:id AND user_id=:uid"),
        {"id": task_id, "uid": user["id"]},
    ).mappings().first()
    if not task:
        raise HTTPException(404, "任務不存在")
    app = _load_application(db, str(task["application_id"]))
    if not app:
        raise HTTPException(404, "投遞不存在")
    payload = build_candidate_payload(db, user["id"])
    skills = ", ".join(s["name"] for s in (payload.get("skills") or []))
    exp_lines = []
    for e in (payload.get("experiences") or [])[:5]:
```

```
        exp_lines.append(
            f"{e.get('start')} ~ {e.get('end')} {e.get('company')} {e.get('title')}: "
            + "; ".join((e.get("bullets") or [])[:2])
        )
    edu_lines = [
        f"{e.get('school')} {e.get('degree')} {e.get('major')}"
        for e in (payload.get("education") or [])
    ]
    fields = {
        "姓名": payload.get("full_name") or "（未填）",
        "手機": payload.get("phone") or "（未填）",
        "郵箱": payload.get("email") or "（未填）",
        "簽證狀態": payload.get("visa_status") or "（未填）",
        "期望薪資": payload.get("salary_expectation") or "（未填）",
        "可到崗時間": payload.get("availability") or "（未填）",
        "技能（複製到技能欄）": skills,
        "工作經歷（最多 5 段，每段兩行內）": "\n".join(exp_lines) or "（未填）",
        "教育（複製到教育欄）": "; ".join(edu_lines) or "（未填）",
    }
    text_block = "\n".join(f"{k}: {v}" for k, v in fields.items())
    # 機器可讀字段 + 簡歷文件（供下載後粘貼或上傳）
    machine_fields = [
        {"key": "name", "label": "姓名", "value": payload.get("full_name") or ""},
        {"key": "phone", "label": "手機", "value": payload.get("phone") or ""},
        {"key": "email", "label": "郵箱", "value": payload.get("email") or ""},
        {"key": "visa", "label": "簽證", "value": payload.get("visa_status") or ""},
        {"key": "salary", "label": "薪資", "value": payload.get("salary_expectation") or ""},
        {"key": "availability", "label": "到崗", "value": payload.get("availability") or ""},
        {"key": "skills", "label": "技能", "value": skills},
        {"key": "experience", "label": "經歷", "value": "\n".join(exp_lines)},
        {"key": "education", "label": "教育", "value": "; ".join(edu_lines)},
    ]
    resume = None
    rf = payload.get("resume_file")
    if rf:
        from urllib.parse import unquote

        from app.core.r2 import object_url

        url = object_url(rf.get("storage_path"))
        resume = {"name": unquote(rf.get("name") or "resume.pdf"), "storage_path":
rf.get("storage_path"),
                  "download_url": url}
    return {"fields": fields, "copy_text": text_block, "machine_fields": machine_fields,
            "resume": resume, "company": app["company_name"], "title": app["title"],
            "apply_url": app.get("apply_url")}


@router.get("")
def list_tasks(user: CurrentUser, db: DB, status: str = "open") -> list[dict]:
    rows = db.execute(
```

```
        text(
            """
            SELECT t.id, t.task_type, t.question, t.options, t.status, t.created_at,
                   t.expires_at, a.status AS app_status,
                   p.company_name, p.title
            FROM manual_tasks t
            JOIN applications a ON a.id = t.application_id
            JOIN job_postings p ON p.id = a.posting_id
            WHERE t.user_id = :uid AND t.status = :st
            ORDER BY t.created_at DESC LIMIT 100
            """
        ),
        {"uid": user["id"], "st": status},
    ).mappings().all()
    return [
        {
            "id": str(r["id"]),
            "task_type": r["task_type"],
            "question": r["question"],
            "options": r["options"] or {},
            "status": r["status"],
            "created_at": r["created_at"].isoformat(),
            "expires_at": r["expires_at"].isoformat(),
            "application_status": r["app_status"],
            "company_name": r["company_name"],
            "title": r["title"],
        }
        for r in rows
    ]


@router.post("/{task_id}/answer", status_code=200)
def answer_task(user: CurrentUser, db: DB, task_id: str, payload: TaskAnswer) -> dict:
    result = answer_and_resume(db, task_id, user["id"], payload.answer)
    if result.get("error"):
        raise HTTPException(404, result["error"])
    return result


@router.post("/{task_id}/cancel", status_code=200)
def cancel_task(user: CurrentUser, db: DB, task_id: str) -> dict:
    row = db.execute(
        text(
            "UPDATE manual_tasks SET status='cancelled' "
            "WHERE id=:id AND user_id=:uid AND status='open' RETURNING application_id"
        ),
        {"id": task_id, "uid": user["id"]},
    ).mappings().first()
    if not row:
        raise HTTPException(404, "任務不存在或已處理")
```

```
    db.execute(
        text(
            "UPDATE applications SET status='aborted', status_reason='人工任務已取消',
updated_at=now() "
            "WHERE id=:aid"
        ),
        {"aid": row["application_id"]},
    )
    db.commit()
    return {"ok": True}
"""用戶認證：驗證前端傳來的 Supabase JWT 並注入當前用戶。

- 前端登錄/註冊走後端認證代理（app/api/auth.py），瀏覽器不直連 supabase.co
- 後端驗證 token：優先調 Supabase /auth/v1/user（權威），失敗退 JWKS 本地驗簽
- 進程內緩存驗證結果（5 分鐘），降低每個請求的網絡往返
- 首次訪問自動補建 profiles 行；ADMIN_EMAILS 中的郵箱自動提升 admin
"""
from __future__ import annotations

import logging
import time
from functools import lru_cache
from typing import Annotated

import httpx
import jwt
from fastapi import Depends, HTTPException, status
from fastapi.security import HTTPAuthorizationCredentials, HTTPBearer
from sqlalchemy import text
from sqlalchemy.orm import Session

from app.config import settings
from app.db import get_db

logger = logging.getLogger(__name__)

_bearer = HTTPBearer(auto_error=False)

# token -> (expire_ts, user_id, email)
_verify_cache: dict[str, tuple[float, str, str | None]] = {}
_CACHE_TTL = 300


@lru_cache(maxsize=1)
def _jwks() -> dict:
    url = f"{settings.supabase_url}/auth/v1/.well-known/jwks.json"
    try:
        resp = httpx.get(url, timeout=10)
        resp.raise_for_status()
        return resp.json()
    except httpx.HTTPError as exc:  # pragma: no cover
```

```
        logger.error("無法獲取 Supabase JWKS: %s", exc)
        raise RuntimeError("Supabase JWKS 不可用") from exc


def _verify_via_api(token: str) -> tuple[str, str | None] | None:
    """調 Supabase /auth/v1/user 驗證，返回 (user_id, email) 或 None。"""
    try:
        resp = httpx.get(
            f"{settings.supabase_url}/auth/v1/user",
            headers={
                "apikey": settings.supabase_anon_key,
                "Authorization": f"Bearer {token}",
            },
            timeout=10,
        )
        if resp.status_code == 200:
            data = resp.json()
            return data["id"], data.get("email")
    except httpx.HTTPError:
        pass
    return None


def _verify_local(token: str) -> tuple[str, str | None] | None:
    try:
        unverified = jwt.get_unverified_header(token)
        kid = unverified["kid"]
        key = next(k for k in _jwks()["keys"] if k["kid"] == kid)
        payload = jwt.decode(
            token,
            key=jwt.algorithms.RSAAlgorithm.from_jwk(key),
            algorithms=["RS256"],
            audience="authenticated",
            options={"verify_aud": True},
        )
        return payload.get("sub"), payload.get("email")
    except Exception:  # noqa: BLE001
        return None


def verify_token(token: str) -> tuple[str, str | None]:
    """驗證 token，返回 (user_id, email)。兩種方式都失敗 → 401。"""
    now = time.time()
    hit = _verify_cache.get(token)
    if hit and hit[0] > now:
        return hit[1], hit[2]
    result = _verify_via_api(token) or _verify_local(token)
    if not result:
        raise HTTPException(status.HTTP_401_UNAUTHORIZED, "登錄已失效，請重新登錄")
    try:
```

```
        exp = jwt.decode(token, options={"verify_signature": False}).get("exp")
        ttl = min(_CACHE_TTL, max(0, int(exp) - int(now))) if exp else _CACHE_TTL
    except Exception:  # noqa: BLE001
        ttl = _CACHE_TTL
    _verify_cache[token] = (now + ttl, result[0], result[1])
    return result


def get_current_user(
    credentials: Annotated[HTTPAuthorizationCredentials | None, Depends(_bearer)],
    db: Annotated[Session, Depends(get_db)],
) -> dict:
    """返回 {'id': uuid, 'email': str|None, 'phone': str|None, 'role': str}。"""
    if credentials is None:
        raise HTTPException(status.HTTP_401_UNAUTHORIZED, "缺少登錄憑證")
    user_id, email = verify_token(credentials.credentials)

    row = db.execute(
        text(
            """
            INSERT INTO profiles (id, email, phone)
            VALUES (:id, :email, :phone)
            ON CONFLICT (id) DO UPDATE
              SET email = COALESCE(profiles.email, EXCLUDED.email),
                  phone = COALESCE(profiles.phone, EXCLUDED.phone)
            RETURNING id::text, email, phone, role, full_name
            """
        ),
        {"id": user_id, "email": email, "phone": None},
    ).mappings().one()
    db.commit()
    result = dict(row)
    # 管理員自動提升：ADMIN_EMAILS 中的郵箱登錄即成為 admin
    email_lower = (result.get("email") or "").lower()
    if email_lower and email_lower in settings.admin_email_list and result.get("role") !=
"admin":
        db.execute(text("UPDATE profiles SET role='admin' WHERE id=:id"), {"id": user_id})
        db.commit()
        result["role"] = "admin"
    return result
"""信封加密（Envelope Encryption）——第三方平台帳密與登錄態的安全存儲。

方案（對應《01-架構設計》第 5 節）：
- 每用戶一把 DEK（Data Encryption Key，AES-256-GCM 隨機生成）
- DEK 用服務端 KEK（環境變量，不入庫）包裹後存 key_wraps 表
- 明文只存在於投遞 Worker 進程內存，永不落庫、不落日誌
"""
from __future__ import annotations

import base64
import os
```

```
from cryptography.hazmat.primitives.ciphers.aead import AESGCM

from app.config import settings

_NONCE_BYTES = 12


def _kek() -> bytes:
    raw = base64.b64decode(settings.kek_base64)
    if len(raw) != 32:
        raise RuntimeError("KEK_BASE64 必須為 32 字節的 base64（openssl rand -base64 32）")
    return raw


def _b64e(data: bytes) -> str:
    return base64.b64encode(data).decode("ascii")


def _b64d(data: str) -> bytes:
    return base64.b64decode(data.encode("ascii"))


def generate_dek() -> bytes:
    """生成用戶 DEK（32 字節）。"""
    return os.urandom(32)


def wrap_dek(dek: bytes) -> str:
    """用 KEK 包裹 DEK 後存庫。"""
    nonce = os.urandom(_NONCE_BYTES)
    ciphertext = AESGCM(_kek()).encrypt(nonce, dek, None)
    return _b64e(nonce + ciphertext)


def unwrap_dek(wrapped: str) -> bytes:
    """解包 DEK（僅在需要解密用戶數據時調用）。"""
    blob = _b64d(wrapped)
    nonce, ciphertext = blob[:_NONCE_BYTES], blob[_NONCE_BYTES:]
    return AESGCM(_kek()).decrypt(nonce, ciphertext, None)


def encrypt_field(dek: bytes, plaintext: str) -> str:
    """用用戶 DEK 加密單個字段，輸出 `nonce:ciphertext` base64。"""
    nonce = os.urandom(_NONCE_BYTES)
    ciphertext = AESGCM(dek).encrypt(nonce, plaintext.encode("utf-8"), None)
    return _b64e(nonce) + ":" + _b64e(ciphertext)


def decrypt_field(dek: bytes, stored: str) -> str:
```

```
    nonce_b64, ciphertext_b64 = stored.split(":", 1)
    return AESGCM(dek).decrypt(_b64d(nonce_b64), _b64d(ciphertext_b64), None).decode("utf-8")


def encrypt_json(dek: bytes, obj: dict) -> str:
    """加密 JSON 結構（如登錄態 Cookie 快照）。"""
    import json

    return encrypt_field(dek, json.dumps(obj, ensure_ascii=False))


def decrypt_json(dek: bytes, stored: str) -> dict:
    import json

    return json.loads(decrypt_field(dek, stored))


def get_dek_for_user(db, user_id: str) -> bytes:
    """取用戶 DEK；不存在則生成並包裹存庫（key_wraps）。"""
    from sqlalchemy import text

    row = db.execute(
        text(
            "SELECT wrapped_dek FROM key_wraps "
            "WHERE owner_type = 'user' AND owner_id = :uid"
        ),
        {"uid": user_id},
    ).mappings().first()
    if row:
        return unwrap_dek(row["wrapped_dek"])
    dek = generate_dek()
    db.execute(
        text(
            "INSERT INTO key_wraps (owner_type, owner_id, wrapped_dek) "
            "VALUES ('user', :uid, :wrapped)"
        ),
        {"uid": user_id, "wrapped": wrap_dek(dek)},
    )
    db.commit()
    return dek
"""審計日誌工具（不影響主流程，失敗僅告警）。"""
from __future__ import annotations

import logging

from sqlalchemy import text
from sqlalchemy.orm import Session

from app.core.jsonutil import j
```

```
logger = logging.getLogger(__name__)


def log_audit(
    db: Session,
    actor_user_id: str,
    action: str,
    entity: str | None = None,
    entity_id: str | None = None,
    meta: dict | None = None,
    ip: str | None = None,
) -> None:
    try:
        db.execute(
            text(
                """
                INSERT INTO audit_logs (actor_user_id, action, entity, entity_id, meta, ip)
                VALUES (:a, :action, :e, :eid, CAST(:meta AS jsonb), :ip)
                """
            ),
            {
                "a": actor_user_id,
                "action": action,
                "e": entity,
                "eid": entity_id,
                "meta": j(meta or {}),
                "ip": ip,
            },
        )
    except Exception:  # noqa: BLE001
        logger.exception("審計日誌寫入失敗")
"""R2 對象存儲客戶端（S3 兼容）。"""
from __future__ import annotations

import boto3
from botocore.config import Config

from app.config import settings

_client = None


def r2_client():
    global _client
    if _client is None:
        _client = boto3.client(
            "s3",
            endpoint_url=f"https://{settings.r2_account_id}.r2.cloudflarestorage.com",
            aws_access_key_id=settings.r2_access_key_id,
            aws_secret_access_key=settings.r2_secret_access_key,
```

```
            region_name="auto",
            config=Config(signature_version="s3v4"),
        )
    return _client


def upload_bytes(key: str, data: bytes, mime: str) -> None:
    r2_client().put_object(
        Bucket=settings.r2_bucket,
        Key=key,
        Body=data,
        ContentType=mime,
    )


def delete_object(key: str) -> None:
    r2_client().delete_object(Bucket=settings.r2_bucket, Key=key)


def object_url(key: str) -> str:
    if settings.r2_public_base_url:
        return f"{settings.r2_public_base_url.rstrip('/')}/{key}"
    return key


def presigned_url(key: str, expires: int = 3600) -> str:
    """私有桶的臨時訪問鏈接（無需公開桶）。"""
    return r2_client().generate_presigned_url(
        "get_object",
        Params={"Bucket": settings.r2_bucket, "Key": key},
        ExpiresIn=expires,
    )
"""LLM 客戶端（OpenAI 兼容，可切換 DeepSeek/OpenAI 等）。

- chat: 普通對話補全
- chat_json: 要求 JSON 輸出並容錯解析（處理 markdown 代碼塊、前後綴雜訊）
- 記錄 token 用量到 ai_jobs（成本控制，GAP-11）
"""
from __future__ import annotations

import json
import logging
import re

from openai import OpenAI

from app.config import settings

logger = logging.getLogger(__name__)
```

```
_client: OpenAI | None = None

# 進程內用量計數（免費層單進程部署夠用；多進程時以 ai_jobs 表為準）
USAGE = {"calls": 0, "tokens_in": 0, "tokens_out": 0}

# ── 隱私脫敏（方案 A）：發送 LLM 前對個人敏感信息打碼 ──────────────────────
_PHONE_RE = re.compile(r"(?<!\d)(1[3-9]\d)(\d{4})(\d{4})(?!\d)")
_IDCARD_RE = re.compile(r"(?<!\d)(\d{6})(\d{8})(\d{3}[\dXx])(?!\d)")
_EMAIL_RE = re.compile(r"([A-Za-z0-9._%+-])[A-Za-z0-9._%+-]*@([A-Za-z0-9.-]+\.[A-Za-z]{2,})")


def mask_pii(text: str) -> str:
    """脫敏：手機 189****9707、身份證前 6 後 4 保留中間打碼、郵箱名打碼。"""
    if not text:
        return text
    text = _PHONE_RE.sub(r"\1****\3", text)
    text = _IDCARD_RE.sub(r"\1********\3", text)
    text = _EMAIL_RE.sub(r"\1***@\2", text)
    return text


def _sanitize_messages(messages: list[dict]) -> list[dict]:
    return [{**m, "content": mask_pii(m.get("content") or "")} for m in messages]


def usage() -> dict:
    return dict(USAGE)


def _get_client() -> OpenAI:
    global _client
    if _client is None:
        if not settings.llm_api_key:
            raise RuntimeError("LLM_API_KEY 未配置（參考 backend/.env.example）")
        _client = OpenAI(base_url=settings.llm_base_url, api_key=settings.llm_api_key)
    return _client


def _extract_json(text: str) -> dict:
    """從 LLM 輸出中容錯提取 JSON 對象。"""
    text = text.strip()
    fence = re.search(r"```(?:json)?\s*(\{.*?\})\s*```", text, re.S)
    if fence:
        text = fence.group(1)
    else:
        start, end = text.find("{"), text.rfind("}")
        if start != -1 and end > start:
            text = text[start : end + 1]
    try:
        return json.loads(text)
```

```
    except json.JSONDecodeError:
        # 修復常見問題：去掉尾隨逗號
        fixed = re.sub(r",\s*([}\]])", r"\1", text)
        return json.loads(fixed)


def chat(
    system: str,
    user: str,
    *,
    model: str | None = None,
    max_tokens: int = 4000,
    temperature: float = 0.2,
) -> str:
    resp = _get_client().chat.completions.create(
        model=model or settings.llm_model,
        messages=[
            {"role": "system", "content": system},
            {"role": "user", "content": mask_pii(user)},
        ],
        max_tokens=max_tokens,
        temperature=temperature,
    )
    if resp.usage:
        USAGE["calls"] += 1
        USAGE["tokens_in"] += resp.usage.prompt_tokens or 0
        USAGE["tokens_out"] += resp.usage.completion_tokens or 0
    return resp.choices[0].message.content or ""


def chat_json(
    system: str,
    user: str,
    *,
    model: str | None = None,
    max_tokens: int = 4000,
    retries: int = 2,
) -> dict:
    """要求 LLM 返回 JSON；解析失敗時自動重試（帶錯誤提示）。"""
    last_err = ""
    for attempt in range(retries + 1):
        prompt = user
        if last_err:
            prompt += f"\n\n【上一次輸出無法解析為 JSON：{last_err}。請只輸出一個合法的 JSON 對象，不要包含
任何其他文字。】"
        try:
            out = chat(
                system + "\n 你必須只輸出一個合法 JSON 對象，不要 markdown 代碼塊，不要任何解釋。",
                prompt,
                model=model or settings.llm_structured_model,
                max_tokens=max_tokens,
```

```
                temperature=0.1,
            )
            return _extract_json(out)
        except (json.JSONDecodeError, ValueError) as exc:
            last_err = str(exc)
            logger.warning("JSON 解析失敗，重試 %s/%s: %s", attempt + 1, retries, exc)
    raise RuntimeError(f"LLM 多次輸出非法 JSON: {last_err}")


def chat_with_tools(
    system: str,
    messages: list[dict],
    tools: list[dict],
    tool_executor,
    *,
    model: str | None = None,
    max_iters: int = 6,
) -> str:
    """函數調用（tool calling）代理循環：LLM 決定調用工具 → 執行 → 回填結果。

    tool_executor: callable(name: str, args: dict) -> dict
  返回最終的助手文本回覆。
    """
    msgs: list[dict] = [{"role": "system", "content": system}, *(_sanitize_messages(messages))]
    for _ in range(max_iters):
        resp = _get_client().chat.completions.create(
            model=model or settings.llm_model,
            messages=msgs,
            tools=tools,
            max_tokens=2000,
            temperature=0.3,
        )
        msg = resp.choices[0].message
        if not msg.tool_calls:
            return msg.content or ""
        msgs.append(msg)  # assistant 帶 tool_calls 的消息
        for tc in msg.tool_calls:
            name = tc.function.name
            try:
                args = json.loads(tc.function.arguments or "{}")
            except json.JSONDecodeError:
                args = {}
            try:
                result = tool_executor(name, args)
            except Exception as exc:  # noqa: BLE001
                logger.exception("工具執行失敗 %s", name)
                result = {"error": str(exc)[:500]}
            msgs.append({
                "role": "tool",
                "tool_call_id": tc.id,
```

```
                "content": json.dumps(result, ensure_ascii=False)[:6000],
            })
    return "（操作步驟過多，請簡化您的問題後再試）"
"""後台調度器：每日採集 + 心跳保活（FR-3a2/3a3，GAP-9 可靠性）。

- 每日 02:00（Asia/Shanghai）執行 collect_all_users
- 每 6 小時心跳：真實查庫，防止 Supabase Free 因 1 週無活動被休眠
- 單進程模型：與 API 同進程（免費層部署），用 collect_jobs 表做任務留痕
"""
from __future__ import annotations

import logging

from apscheduler.schedulers.background import BackgroundScheduler
from apscheduler.triggers.cron import CronTrigger
from apscheduler.triggers.interval import IntervalTrigger
from sqlalchemy import text

from app.collectors.pipeline import collect_all_users
from app.core.jsonutil import j
from app.db import get_engine

logger = logging.getLogger(__name__)

_scheduler: BackgroundScheduler | None = None


def _job_collect() -> None:
    from app.db import SessionLocal  # 惰性，避免無 DB 配置時啟動失敗

    logger.info("每日採集任務開始")
    db = SessionLocal()
    try:
        db.execute(
            text(
                "INSERT INTO collect_jobs (kind, status, scheduled_at) "
                "VALUES ('collect', 'running', now()) RETURNING id"
            )
        )
        report = collect_all_users(db)
        db.execute(
            text(
                "UPDATE collect_jobs SET status='done', finished_at=now(), payload=CAST(:p AS
jsonb) "
                "WHERE kind='collect' AND status='running' AND finished_at IS NULL"
            ),
            {"p": j(report)},
        )
        db.commit()
        logger.info("每日採集完成: %s", report)
    except Exception:  # noqa: BLE001
```

```
        logger.exception("每日採集失敗")
        try:
            db.execute(
                text(
                    "UPDATE collect_jobs SET status='failed', finished_at=now() "
                    "WHERE kind='collect' AND status='running' AND finished_at IS NULL"
                )
            )
            db.commit()
        except Exception:  # noqa: BLE001
            pass
    finally:
        db.close()


def _job_heartbeat() -> None:
    try:
        with get_engine().connect() as conn:
            conn.execute(text("SELECT 1"))
        logger.debug("心跳 OK")
    except Exception as exc:  # noqa: BLE001
        logger.warning("心跳失敗（數據庫未配置或不可達）: %s", exc)


def _job_daily_push() -> None:
    """每 10 分鐘掃描：到達用戶設定推送時間（本地時區）且當日未發 → 發每日摘要。"""
    from app.db import SessionLocal
    from app.services.digest import deliver_daily_digest

    db = SessionLocal()
    try:
        rows = db.execute(
            text(
                """
                SELECT user_id FROM user_settings s
                WHERE s.daily_digest_enabled = true
                  AND to_char(now() AT TIME ZONE s.timezone, 'HH24:MI') =
to_char(s.push_time_local, 'HH24:MI')
                  AND NOT EXISTS (
                    SELECT 1 FROM notifications n
                    WHERE n.user_id = s.user_id AND n.kind = 'daily_digest'
                      AND n.sent_at >= (now() AT TIME ZONE s.timezone)::date
                  )
                """
            )
        ).mappings().all()
        for r in rows:
            try:
                deliver_daily_digest(db, str(r["user_id"]))
            except Exception:  # noqa: BLE001
                logger.exception("每日摘要推送失敗 user=%s", r["user_id"])
```

```
    finally:
        db.close()


def _job_apply_queue() -> None:
    """每 30 分鐘：處理到期投遞隊列（queued + 到期重試）。"""
    from app.db import SessionLocal
    from app.services.apply_engine import process_queue

    db = SessionLocal()
    try:
        report = process_queue(db)
        if report["processed"]:
            logger.info("投遞隊列處理：%s", report)
    except Exception:  # noqa: BLE001
        logger.exception("投遞隊列處理失敗")
    finally:
        db.close()


def start_scheduler() -> BackgroundScheduler:
    global _scheduler
    if _scheduler is not None:
        return _scheduler
    _scheduler = BackgroundScheduler(timezone="Asia/Shanghai")
    _scheduler.add_job(_job_collect, CronTrigger(hour=2, minute=0), id="daily_collect",
                       misfire_grace_time=3600, coalesce=True, max_instances=1)
    _scheduler.add_job(_job_daily_push, IntervalTrigger(minutes=10), id="daily_push",
                       coalesce=True, max_instances=1)
    _scheduler.add_job(_job_apply_queue, IntervalTrigger(minutes=30), id="apply_queue",
                       coalesce=True, max_instances=1)
    _scheduler.add_job(_job_heartbeat, IntervalTrigger(hours=6), id="heartbeat",
                       coalesce=True, max_instances=1)
    _scheduler.start()
    logger.info("調度器已啟動（每日採集 02:00 + 摘要推送 10min + 投遞隊列 30min + 心跳 6h）")
    return _scheduler


def stop_scheduler() -> None:
    global _scheduler
    if _scheduler is not None:
        _scheduler.shutdown(wait=False)
        _scheduler = None
"""通用投遞器（本系統唯一的投遞適配器）。

策略：打開官網申請頁 → 收集可見輸入框 → 按 name/placeholder/標籤關鍵詞映射到候選人字段
→ 填充（缺的從個人信息庫按相似度互補）→ 上傳簡歷 → 嘗試提交 → 檢測成功信號。

原則：
- 不按平台定制：所有網站走同一套字段匹配，站點改版不影響邏輯
```

```
- 不猜結果：無法確認提交成功時一律生成「待確認」人工任務，由用戶在官網核對後勾選確認
- 卡點轉人工：驗證碼、必填缺數據、找不到提交按鈕，全部轉人工接管，不硬來、不編造
"""
from __future__ import annotations

import json
import logging

from app.apply.base import ApplyContext, ApplyResult, ManualRequest, register_adapter
from app.apply.browser import (
    detect_captcha,
    human_pause,
    new_stealth_context,
    take_screenshot,
    upload_file,
)

logger = logging.getLogger(__name__)

# 表單字段關鍵詞 → payload 鍵
FIELD_MAP = [
    (("first", "given"), None, "first_name"),
    (("last", "family", "surname"), None, "last_name"),
    (("email",), None, "email"),
    (("phone", "mobile", "tel"), None, "phone"),
    (("city", "location"), None, "location"),
    (("salary", "expected pay", "期望薪资"), None, "salary_expectation"),
    (("available", "start date", "到岗"), None, "availability"),
    (("visa", "work authorization", "签证"), None, "visa_status"),
    (("linkedin",), None, "linkedin_url"),
    (("website", "portfolio", "github"), None, "portfolio_url"),
    (("cover_letter", "cover letter", "coverletter", "求职信"), None, "cover_letter"),
    (("name",), None, "full_name"),
]

SUCCESS_HINTS = ("thank you", "application submitted", "application received", "successfully",
"已提交", "提交成功")


@register_adapter
class UniversalApplyAdapter:
    """通用投遞器：所有招聘來源共用。"""

    name = "generic"

    def apply(self, ctx: ApplyContext) -> ApplyResult:
        try:
            from playwright.sync_api import sync_playwright
        except ImportError as exc:  # pragma: no cover
            return ApplyResult(False, error_code="playwright_missing", error_detail=str(exc))
```

```
        p = ctx.payload
        shots: list[str] = []

        with sync_playwright() as pw:
            browser, page_ctx = new_stealth_context(pw)
            page = page_ctx.new_page()
            try:
                page.goto(ctx.posting["apply_url"], wait_until="domcontentloaded",
timeout=45000)
                page.wait_for_timeout(3500)
                # 多步表單：先嘗試點擊「Apply」進入申請流程
                for text in ("Apply", "Apply Now", "申请", "立即申请"):
                    try:
                        btn = page.locator(f"button:has-text('{text}'), a:has-
text('{text}')").first
                        if btn.count() and btn.is_visible():
                            btn.click()
                            page.wait_for_load_state("domcontentloaded")
                            page.wait_for_timeout(3000)
                            break
                    except Exception:  # noqa: BLE001
                        continue

                if detect_captcha(page):
                    shot = take_screenshot(page, "captcha.png", _shot_dir(ctx.user_id))
                    if shot:
                        shots.append(shot)
                    browser.close()
                    return ApplyResult(
                        False, error_code="captcha",
                        manual=ManualRequest("captcha", "申請頁出現人機驗證，請處理後點擊「已處理」",
                                             {"confirm": {"label": "我已處理驗證碼", "type":
"checkbox"}}, shot),
                        screenshots=shots,
                    )

                inputs = _collect_inputs(page)
                unfilled_required: list[str] = []
                auto_fill_log: list[str] = []
                for inp in inputs:
                    if inp["type"] == "file":
                        continue
                    value, matched = _value_for(inp, p, ctx)
                    # 元信息表互補：簡歷/畫像沒有 → 查個人信息庫（相似度 ⩾50% 自動填）
                    if not value and inp["editable"]:
                        from app.api.info_fields import resolve_field
                        from app.db import SessionLocal

                        field_label = inp["label"] or inp["name"] or inp["placeholder"]
                        if field_label:
                            try:
                                s = SessionLocal()
                                try:
```

```
                    filled, src_key, score = resolve_field(
                        s, ctx.user_id, inp["name"], field_label)
                finally:
                    s.close()
                if filled:
                    value = filled
                    auto_fill_log.append(
                        f"{field_label} ← 信息庫[{src_key}] 相似度{score}")
            except Exception:  # noqa: BLE001
                pass
    if value and inp["editable"]:
        # 官網格子校驗適配：字符上限截斷、數字範圍裁剪，不卡住
        raw = value
        if inp.get("maxlength") and len(value) > inp["maxlength"]:
            value = value[: inp["maxlength"]]
        if inp.get("type") == "number" and inp.get("max") is not None:
            try:
                if float(value) > float(inp["max"]):
                    value = str(inp["max"])
            except (TypeError, ValueError):  # noqa: BLE001
                pass
        if value != raw:
            auto_fill_log.append(
                f"{inp.get('label') or inp.get('name')}: 按官網要求截斷/調整 "
                f"({len(raw)}→{len(value)} 字符)")
        try:
            page.locator(inp["selector"]).first.fill(value)
        except Exception:  # noqa: BLE001
            pass
    elif matched and inp["required"] and inp["editable"] and not value:
        unfilled_required.append(inp["label"] or inp["name"] or inp["selector"])
human_pause()

if ctx.resume_local_path and any(i["type"] == "file" for i in inputs):
    upload_file(page, ["input[type=file]"], ctx.resume_local_path)
    human_pause(1.0, 2.5)

shot = take_screenshot(page, "before_submit.png", _shot_dir(ctx.user_id))
if shot:
    shots.append(shot)

if unfilled_required:
    browser.close()
    return ApplyResult(
        False, error_code="missing_field",
        error_detail="必填字段缺數據: " + ", ".join(unfilled_required[:6]),
        manual=ManualRequest(
            "missing_field",
            "申請表有必填字段沒有對應數據，請補充（格式：每行「字段=值」）",
            {"values": {"label": f"待填字段: {', '.join(unfilled_required[:6])}",
```

```
                                    "type": "textarea"}},
                            shot,
                        ),
                        screenshots=shots,
                    )

                if ctx.dry_run:
                    browser.close()
                    return ApplyResult(True, external_ref="DRY_RUN", screenshots=shots)

                submitted = False
                for sel in ("button[type=submit]", "input[type=submit]",
                            "button:has-text('Submit')", "button:has-text('提交')"):
                    loc = page.locator(sel).first
                    if loc.count() and loc.is_visible():
                        loc.click()
                        page.wait_for_timeout(6000)
                        submitted = True
                        break
                content = (page.content() or "").lower()
                success = submitted and any(h in content for h in SUCCESS_HINTS)
                shot = take_screenshot(page, "after_submit.png", _shot_dir(ctx.user_id))
                if shot:
                    shots.append(shot)
                browser.close()
                if success:
                    return ApplyResult(True, external_ref=ctx.posting["external_id"],
screenshots=shots)
                return ApplyResult(
                    False, error_code="submit_unconfirmed" if submitted else "no_submit_button",
                    error_detail="提交未確認成功，請人工確認",
                    manual=ManualRequest(
                        "confirm_content",
                        "表單已填寫並嘗試提交，但未能確認結果。請打開官網連結核對，"
                        "提交成功後勾選確認並填入申請編號。",
                        {"note": {"label": "補充說明（可空）", "type": "text"},
                         "confirmed": {"label": "已確認提交成功", "type": "checkbox"}}, shot,
                    ),
                    screenshots=shots,
                )
            except Exception as exc:  # noqa: BLE001
                logger.exception("通用投遞異常 app=%s", ctx.application_id)
                try:
                    browser.close()
                except Exception:  # noqa: BLE001
                    pass
                return ApplyResult(False, error_code="adapter_error",
                                   error_detail=str(exc)[:800], screenshots=shots)


def _collect_inputs(page) -> list[dict]:
```

```
    return page.evaluate(
        """
        () => Array.from(document.querySelectorAll(
          'input:not([type=hidden]), textarea, select'
        )).map((el, i) => {
          const label = el.labels && el.labels[0] ? el.labels[0].innerText.trim()
            : (el.closest('label') ? el.closest('label').innerText.trim() : '');
          const id = 'jobfly_field_' + i;
          if (!el.id) el.id = id;
          const hint = (el.closest('.form-item, .el-form-item, div, label') || el)
            ? ((el.closest('.el-form-item') || el.closest('label') || el.parentElement) ||
{}).innerText || ''
            : '';
          return {
            selector: '#' + el.id,
            type: (el.type || el.tagName.toLowerCase()),
            name: el.name || '',
            placeholder: el.placeholder || '',
            label: label,
            hint: hint.trim().slice(0, 300),
            required: el.required || el.getAttribute('aria-required') === 'true',
            editable: !(el.readOnly || el.disabled),
            tag: el.tagName.toLowerCase(),
            maxlength: el.maxLength > 0 ? el.maxLength : null,
            pattern: el.pattern || null,
            min: el.min || null,
            max: el.max || null,
          };
        })
        """
    )


def _value_for(inp: dict, payload: dict, ctx: ApplyContext) -> tuple[str, bool]:
    hay = f"{inp['name']} {inp['placeholder']} {inp['label']}".lower()
    for keywords, _, field in FIELD_MAP:
        if any(k in hay for k in keywords):
            if field == "first_name":
                return _split_name(payload.get("full_name") or "")[0], True
            if field == "last_name":
                return _split_name(payload.get("full_name") or "")[1], True
            if field == "cover_letter":
                return ctx.cover_letter or "", True
            if field in ("linkedin_url", "portfolio_url"):
                v = payload.get(field)
                return v or "", bool(v)
            v = payload.get(field)
            if isinstance(v, (dict, list)):
                v = ""
            return str(v or ""), True
    return "", False
```

```
def _split_name(full: str) -> tuple[str, str]:
    parts = (full or "Candidate").strip().split()
    if len(parts) >= 2:
        return parts[0], " ".join(parts[1:])
    return parts[0] if parts else "Candidate", "."


def _shot_dir(user_id: str) -> str:
    import os
    import tempfile

    d = os.path.join(tempfile.gettempdir(), "jobfly-shots", user_id)
    os.makedirs(d, exist_ok=True)
    return d
"""貼合度匹配服務（FR-1f，GAP-12 雙層評分）。

第一層：硬性條件逐項核對（學歷/年限/簽證/語言/證書/地點），任一不滿足 → hard_failed。
第二層：LLM 綜合評分（50-100）+ 分項 + 可解釋理由。
結果 upsert 到 job_matches；status 由 auto_apply_mode 在投遞階段消費。
"""
from __future__ import annotations

from sqlalchemy import text
from sqlalchemy.orm import Session

from app.core.jsonutil import j
from app.core.llm import chat_json

_HARD_SYSTEM = """
你是一名招聘顧問。從 JD 中抽取硬性要求。
規則：
1. kind 取值：education(學歷/專業)、experience_years(工作年限)、visa(簽證/身份)、language(語言)、
certificate(證書)、location(工作地點/在場要求)、other。
2. must=true 表示"必須滿足"類要求（如"要求本科以上"、"需持綠卡"）；偏好類要求 must=false。
3. requirement 用一句話描述。
輸出 JSON（不要輸出其他內容）：
{"requirements": [{"field": "字段名", "kind": "...", "requirement": "描述", "must": true}]}
"""

_SCORE_SYSTEM = """
你是一名招聘顧問。綜合評估候選人與 JD 的貼合度，輸出 50-100 分。
規則：
1. 評分依據：技能重疊、經歷相關性、職級匹配、地點、薪資預期。
2. 若 JD 有硬性要求候選人不滿足，不要評分（此場景由系統先過濾，你不會收到）。
3. 給出三條簡潔可執行的理由（如"缺 XX 技能，可補充學習"）。
輸出 JSON（不要輸出其他內容）：
{"match_score": 50-100, "breakdown": {"skills_overlap": 0-100, "experience_fit": 0-100,
"level_fit": 0-100, "location_fit": 0-100, "salary_fit": 0-100}, "reasons": ["理由 1", "理由 2",
"理由 3"]}
"""
```

```
_SEMANTIC_SYSTEM = """
你是招聘篩選助手。候選人的資料是結構化字段 + 個人信息庫問答的合集。
規則檢查可能因為字段命名差異誤判，你需要用「聯想能力」判斷：
候選人資料中的任何一條信息（無論放在哪個字段）只要能**實質滿足**該要求，就判 pass=true。
例如：要求「熟練使用中英文作為工作語言」，信息庫寫「英文：專業（可作為工作語言）」→ pass=true。
要求「碩士學歷」而教育經歷最高為本科 → pass=false（不要放水）。
輸出 JSON（不要輸出其他內容）：
{"results": [{"index": 0, "pass": true, "note": "一句話原因"}]}
"""


def extract_hard_requirements(jd_text: str) -> list[dict]:
    out = chat_json(_HARD_SYSTEM, f"JD: \n{jd_text[:8000]}")
    return [r for r in (out.get("requirements") or []) if r.get("must")]


def _check_against(profile: dict, req: dict) -> bool:
    """簡化的規則核對 + 兜底。返回是否通過。"""
    kind = req.get("kind")
    requirement = (req.get("requirement") or "").lower()
    hard = profile.get("hard_facts") or {}

    if kind == "experience_years":
        years = hard.get("total_years")
        if years is None:
            return True  # 無法核對不攔截，交由 LLM 評分
        import re

        m = re.search(r"(\d+)", requirement)
        return years >= int(m.group(1)) if m else True
    if kind == "visa":
        return bool(hard.get("visa_status"))  # 有簽證信息即放行，細粒度在 LLM 評分
    if kind == "language":
        langs = " ".join(profile.get("languages") or [])
        return bool(langs)  # 有語言信息即放行，具體匹配度在 LLM 評分
    if kind == "certificate":
        certs = " ".join(profile.get("certificates") or [])
        return bool(certs)
    if kind == "location":
        locs = " ".join(hard.get("preferred_locations") or [])
        return bool(locs)
    return True


def _semantic_recheck(db: Session, user_id: str, profile: dict, requirements: list[dict]) ->
list[dict]:
    """聯想能力兜底：規則判定失敗的要求，用 LLM 帶候選人全量資料語義確認。"""
    info = db.execute(
        text("SELECT key, value FROM user_info_fields WHERE user_id=:uid"),
        {"uid": user_id},
    ).mappings().all()
```

```
    candidate = {
        "硬性事實": profile.get("hard_facts") or {},
        "語言能力": profile.get("languages") or [],
        "技能": [s.get("name") if isinstance(s, dict) else s for s in (profile.get("skills") or
[])],
        "教育": profile.get("education") or [],
        "經歷": profile.get("experiences") or [],
        "個人信息庫": {r["key"]: r["value"] for r in info},
    }
    out = chat_json(
        _SEMANTIC_SYSTEM,
        f"候選人資料：{j(candidate)}\n待確認要求：{j(requirements)}",
    )
    results = out.get("results") or []
    by_index = {int(r.get("index", -1)): r for r in results if isinstance(r, dict)}
    return [
        f for i, f in enumerate(requirements)
        if not (by_index.get(i) or {}).get("pass", False)
    ]


def score_posting(db: Session, user_id: str, posting_id: str) -> dict:
    """對單個崗位執行雙層評分並落庫。返回 match 記錄。"""
    profile = db.execute(
        text(
            "SELECT * FROM candidate_profiles WHERE user_id=:uid AND is_current=true "
            "ORDER BY generated_at DESC LIMIT 1"
        ),
        {"uid": user_id},
    ).mappings().first()
    posting = db.execute(
        text("SELECT * FROM job_postings WHERE id=:pid"),
        {"pid": posting_id},
    ).mappings().first()
    if not profile:
        raise ValueError("請先生成用戶畫像")
    if not posting:
        raise ValueError("崗位不存在")

    # 統一信息源：元信息表（含語言能力等問答）合併進畫像，兩邊永不失同步
    info = db.execute(
        text("SELECT key, value FROM user_info_fields WHERE user_id=:uid"),
        {"uid": user_id},
    ).mappings().all()
    info_map = {r["key"]: r["value"] for r in info}
    profile = dict(profile)
    hf = dict(profile.get("hard_facts") or {})
    for k, v in info_map.items():
        if k in ("visa_status", "preferred_locations", "salary_expectation",
                 "availability", "total_years", "candidate_name"):
            hf[k] = v
```

```
    # total_years 統一為數字（信息庫可能存字符串）
    try:
        hf["total_years"] = float(hf.get("total_years") or 0)
    except (TypeError, ValueError):
        hf["total_years"] = 0.0
    profile["hard_facts"] = hf
    # 語言能力: info_fields 的 languages/語言能力 key 轉為列表供硬性條件檢查
    lang_val = info_map.get("languages") or info_map.get("語言能力") or info_map.get("语言能力")
    if lang_val:
        profile["languages"] = [s.strip() for s in str(lang_val).replace("、", ",").split(",") if
s.strip()]
    elif hf.get("languages"):
        profile["languages"] = hf["languages"] if isinstance(hf["languages"], list) else
[str(hf["languages"])]

    hard_reqs = extract_hard_requirements(posting["description_text"] or posting["title"])
    # location 類要求不硬攔截（交給地區加權處理: 命中加分、未命中降分）
    failures = [
        r for r in hard_reqs
        if r.get("kind") != "location" and not _check_against(dict(profile), r)
    ]
    # 聯想能力兜底: 規則失敗的要求交由 LLM 語義二次確認（字段命名差異不再誤殺）
    if failures:
        try:
            failures = _semantic_recheck(db, user_id, dict(profile), failures)
        except Exception:  # noqa: BLE001
            logger.exception("硬性要求語義確認失敗，按規則結果處理 user=%s", user_id)

    # 校招/社招偏好與職級適配（投遞規則 R2 修訂版）:
    # - 用戶偏好只投社招 → 校招崗位硬性過濾
    # - 用戶允許投校招 → 有工作經歷者不攔截，但降權（職級不適配）
    campus_adjust = 0
    st = db.execute(
        text("SELECT campus_preference FROM user_settings WHERE user_id=:u"),
        {"u": user_id},
    ).mappings().first()
    preference = (st["campus_preference"] if st else "both") or "both"
    is_campus = bool(posting.get("is_campus")) or _is_campus_posting(posting)
    years = (dict(profile).get("hard_facts") or {}).get("total_years") or 0
    if is_campus and preference == "society":
        failures.append({
            "field": "campus_preference",
            "kind": "other",
            "requirement": "用戶設置為只投社招，校招崗位已過濾",
            "passed": False,
        })
    elif is_campus and years >= 1:
        campus_adjust = -15  # 校招職級不適配（有工作經歷）降權
    elif (not is_campus) and preference == "campus":
        failures.append({
            "field": "campus_preference",
            "kind": "other",
```

```
            "requirement": "用戶設置為只投校招，社招崗位已過濾",
            "passed": False,
        })

    if failures:
        db.execute(
            text(
                """
                INSERT INTO job_matches (user_id, posting_id, profile_version_id, hard_failures,
status)
                VALUES (:uid, :pid, :pv, CAST(:hf AS jsonb), 'hard_failed')
                ON CONFLICT (user_id, posting_id) DO UPDATE
                  SET hard_failures=EXCLUDED.hard_failures, status='hard_failed',
updated_at=now()
                """
            ),
            {"uid": user_id, "pid": posting_id, "pv": profile["id"], "hf": j(failures)},
        )
        db.commit()
        return {"match_score": None, "hard_failures": failures, "status": "hard_failed"}

    jd = (posting["description_text"] or "")[:8000]
    prompt = (
        f"候選人概覽：{profile['summary']}\n"
        f"技能：{profile['skills']}\n"
        f"經歷：{profile['experiences']}\n"
        f"硬性事實：{profile['hard_facts']}\n\n"
        f"JD：{jd}"
    )

    # ── 崗位方向：逐方向分別評分（投遞仍一崗一投；崗位級分數取最高方向）──
    raw = posting.get("raw") or {}
    if isinstance(raw, str):
        try:
            raw = json.loads(raw)
        except ValueError:
            raw = {}
    directions = [d for d in (raw.get("directions") or []) if d]
    direction_scores: list[dict] = []
    if len(directions) > 1:
        for d in directions:
            try:
                s = chat_json(_SCORE_SYSTEM, prompt + f"\n（本崗位方向：{d}。請只針對該方向評分。）")
                direction_scores.append({
                    "direction": d,
                    "match_score": min(max(int(s["match_score"]), 50), 100),
                    "reasons": s.get("reasons") or [],
                })
            except Exception:  # noqa: BLE001
                logger.warning("方向評分失敗 direction=%s posting=%s", d, posting_id)
        if direction_scores:
            best = max(direction_scores, key=lambda x: x["match_score"])
```

```
            scored = {"match_score": best["match_score"], "breakdown": {},
                      "reasons": best["reasons"] + [f"（按最匹配方向「{best['direction']}」計）"]}
        else:
            scored = chat_json(_SCORE_SYSTEM, prompt)
    else:
        scored = chat_json(_SCORE_SYSTEM, prompt)

    # ── 貼合度 = 純個人條件匹配（LLM），不做任何偏好加減 ──
    fit_score = min(max(int(scored["match_score"]), 50), 100)

    # ── 投遞優先級 = 貼合度 + 偏好加權（地區/校招職級/總部分支），只用於排序與投遞選擇 ──
    loc_adj = _location_adjust(db, user_id, posting["location"])
    org_adj = _org_adjust(db, user_id, posting.get("org_level") or "unknown")
    priority = min(max(fit_score + loc_adj + campus_adjust + org_adj, 50), 100)
    breakdown = scored.get("breakdown") or {}
    breakdown.update({
        "fit_base": fit_score,
        "location_adjust": loc_adj,
        "campus_adjust": campus_adjust,
        "org_adjust": org_adj,
    })

    db.execute(
        text(
            """
            INSERT INTO job_matches
              (user_id, posting_id, profile_version_id, match_score, priority_score,
               score_breakdown, reasons, direction_scores, status)
            VALUES (:uid, :pid, :pv, :score, :prio, CAST(:bd AS jsonb), CAST(:reasons AS jsonb),
                    CAST(:ds AS jsonb), 'new')
            ON CONFLICT (user_id, posting_id) DO UPDATE
              SET match_score=EXCLUDED.match_score, priority_score=EXCLUDED.priority_score,
                  score_breakdown=EXCLUDED.score_breakdown,
                  reasons=EXCLUDED.reasons, direction_scores=EXCLUDED.direction_scores,
                  hard_failures=NULL,
                  status=CASE WHEN job_matches.status IN ('applied','applying') THEN
job_matches.status ELSE 'new' END,
                  updated_at=now()
            """
        ),
        {
            "uid": user_id,
            "pid": posting_id,
            "pv": profile["id"],
            "score": fit_score,
            "prio": priority,
            "bd": j(breakdown),
            "reasons": j(scored.get("reasons") or []),
            "ds": j(direction_scores or []),
        },
    )
```

```
    db.commit()
    return {"match_score": fit_score, "priority_score": priority, "hard_failures": [],
            "status": "new", "location_adjust": loc_adj,
            "direction_scores": direction_scores}


def _org_adjust(db: Session, user_id: str, org_level: str) -> int:
    """總部/分支偏好加權：偏好總部時分支 -8；偏好分支時總部 -5；一視同仁 0。"""
    st = db.execute(
        text("SELECT org_preference FROM user_settings WHERE user_id=:u"),
        {"u": user_id},
    ).mappings().first()
    pref = (st["org_preference"] if st else "both") or "both"
    if pref == "headquarter" and org_level == "branch":
        return -8
    if pref == "branch" and org_level == "headquarter":
        return -5
    return 0


def _is_campus_posting(posting: dict) -> bool:
    """判斷是否校招/應屆類崗位（標題/JD/採集元數據）。"""
    raw = posting.get("raw") or {}
    if isinstance(raw, str):  # 防禦：raw 偶爾以字符串返回
        try:
            import json as _json

            raw = _json.loads(raw)
        except ValueError:
            raw = {}
    text = " ".join([
        posting.get("title") or "",
        posting.get("description_text") or "",
        str(raw.get("tab") or ""),
        str(raw.get("recruit_type") or ""),
    ]).lower()
    hints = ("校招", "校园", "应届", "應屆", "管培生", "培训生", "培訓生", "秋招", "春招", "2027届",
"2027屆")
    return any(h in text for h in hints)


def _location_adjust(db: Session, user_id: str, posting_location: str | None) -> int:
    """期望城市命中 +location_weight；未命中且配置了期望城市 → -min(weight,10)。"""
    st = db.execute(
        text("SELECT preferred_locations, location_weight FROM user_settings WHERE user_id=:u"),
        {"u": user_id},
    ).mappings().first()
    preferred = (st["preferred_locations"] if st else None) or []
    weight = (st["location_weight"] if st else None) or 10
    if not preferred:
        return 0
```

```
    loc = (posting_location or "").lower()
    if any(p.lower() in loc for p in preferred):
        return weight
    return -min(weight, 10)
"""投遞引擎（FR-3b/3c）：狀態機、預檢、文檔準備、適配器調度、重試、人工接管、結果通知。

狀態流轉（對應《01-架構設計》4.3）：
queued → in_progress → submitted / needs_input(→answered→queued) / failed_retryable / aborted
"""
from __future__ import annotations

import logging
import os
import tempfile
from datetime import datetime, timedelta, timezone

from sqlalchemy import text
from sqlalchemy.orm import Session

from app.apply import base as apply_base
from app.config import settings as app_settings
from app.core.audit import log_audit
from app.core.jsonutil import j
from app.services.form_mapping import build_candidate_payload, missing_fields
from app.api.info_fields import save_answer
from app.services.notifier import notify_with_fallback
from app.services.optimize import (
    attach_cover_letter,
    generate_cover_letter,
    optimize_resume_for_jd,
)
from app.services.parsing import download_bytes

logger = logging.getLogger(__name__)

MAX_RETRIES = 3
RETRY_BASE_MINUTES = 30
QUEUE_LIMIT = 50  # 每輪 process_queue 處理上限（免費層資源控制）

# 人工回填允許覆蓋的字段白名單（防止注入任意鍵）
ANSWER_WHITELIST = {
    "full_name", "email", "phone", "location", "visa_status", "availability",
    "salary_expectation", "linkedin_url", "portfolio_url", "sms_code",
}


# ── 1. 隊列生成（FR-1g 三態策略） ──────────────────────────────────────


def create_applications_from_matches(db: Session, user_id: str) -> dict:
    st = db.execute(
```

```
    text(
        "SELECT auto_apply_mode, match_threshold, campus_preference "
        "FROM user_settings WHERE user_id=:uid"
    ),
    {"uid": user_id},
).mappings().first()
mode = st["auto_apply_mode"] if st else "threshold"
if mode == "off":
    return {"mode": "off", "created": 0}
threshold = st["match_threshold"] if st else 70
pref = (st["campus_preference"] if st else "both") or "both"
where = "m.status IN ('new','pushed') AND m.match_score IS NOT NULL"
params: dict = {"uid": user_id}
if mode == "threshold":
    where += " AND m.match_score >= :th"
    params["th"] = threshold
campus_where = ""
if pref == "society":
    campus_where = "AND p.is_campus = false"
elif pref == "campus":
    campus_where = "AND p.is_campus = true"
result = db.execute(
    text(
        f"""
        INSERT INTO applications (user_id, posting_id, match_id, status)
        SELECT m.user_id, m.posting_id, m.id, 'queued'
        FROM job_matches m JOIN job_postings p ON p.id = m.posting_id
        WHERE m.user_id = :uid AND {where} AND p.is_active = true
          {campus_where}
          -- 公司級一崗一投：同一公司已有未終止投遞時跳過其餘崗位
          AND NOT EXISTS (
            SELECT 1 FROM applications a2
            JOIN job_postings p2 ON p2.id = a2.posting_id
            WHERE a2.user_id = :uid AND p2.company_name = p.company_name
              AND a2.status NOT IN ('aborted', 'duplicate_skip', 'not_supported')
          )
        ORDER BY m.priority_score DESC NULLS LAST, m.match_score DESC NULLS LAST
        ON CONFLICT (user_id, posting_id) DO NOTHING
        """
    ),
    params,
)
db.execute(
    text(
        f"""
        UPDATE job_matches SET status='applying', updated_at=now()
        WHERE user_id = :uid AND status IN ('new','pushed')
          AND match_score IS NOT NULL
          {("AND match_score >= :th") if mode == 'threshold' else ''}
        """
```

```
        ),
        params,
    )
    db.commit()
    return {"mode": mode, "created": result.rowcount}


# ── 2. 預檢 ──────────────────────────────────────────────────────────────────


def _preflight(payload: dict) -> dict | None:
    """返回阻塞原因（manual request 描述），無阻塞返回 None。"""
    miss = missing_fields(payload)
    if miss["critical"]:
        return {
            "task_type": "missing_field",
            "question": f"投遞前缺關鍵信息：{', '.join(miss['critical'])}。請補全後重試。",
            "options": {f: {"label": f, "type": "text"} for f in miss["critical"]},
        }
    if not payload.get("resume_file"):
        return {
            "task_type": "missing_file",
            "question": "尚未上傳可用簡歷（需上傳並解析成功）。請先在「我的文件」上傳簡歷。",
            "options": {"confirmed": {"label": "我已上傳簡歷", "type": "checkbox"}},
        }
    return None


# ── 3. 上下文準備（簡歷版本選擇 + AI 微調 + CL + 本地文件）──────────────────────

# 判斷需要英文簡歷的地點/地區關鍵詞（中國大陸以外）
_EN_HINTS = (
    "hong kong", "香港", "singapore", "london", "new york", "san francisco",
    "tokyo", "seoul", "sydney", "toronto", "vancouver", "remote", "海外",
    "united states", "united kingdom", "canada", "australia",
)


def _pick_resume_lang(app: dict) -> str:
    """根據崗位地點/公司選擇簡歷語言：海外/香港 → en，否則 zh。"""
    location = (app.get("location") or "").lower()
    text = f"{location} {(app.get('description_text') or '')[:300]}".lower()
    return "en" if any(h in text for h in _EN_HINTS) else "zh"


def _prepare_context(db: Session, app: dict, payload: dict, settings_row: dict) ->
apply_base.ApplyContext:
    want_lang = _pick_resume_lang(app)
    version = db.execute(
        text(
            "SELECT id, content_md FROM resume_versions WHERE user_id=:uid "
            "AND content_md IS NOT NULL AND COALESCE(lang,'zh')=:lang "
```

```
            "ORDER BY version_no DESC LIMIT 1"
        ),
        {"uid": app["user_id"], "lang": want_lang},
    ).mappings().first()
    if not version:
        # 沒有對應語言的簡歷時回退到最新任意版本
        version = db.execute(
            text(
                "SELECT id, content_md FROM resume_versions WHERE user_id=:uid "
                "AND content_md IS NOT NULL ORDER BY version_no DESC LIMIT 1"
            ),
            {"uid": app["user_id"]},
        ).mappings().first()
    if not version:
        raise RuntimeError("沒有可用的簡歷文本（請先解析簡歷）")
    version_id, resume_text = str(version["id"]), version["content_md"]

    jd = app.get("description_text")
    if settings_row.get("ai_optimize_resume") and jd:
        new_vid = optimize_resume_for_jd(db, app["user_id"], version_id, jd)
        if new_vid:
            new_row = db.execute(
                text("SELECT content_md FROM resume_versions WHERE id=:id"),
                {"id": new_vid},
            ).mappings().first()
            version_id, resume_text = new_vid, new_row["content_md"]
            db.execute(
                text("UPDATE applications SET resume_version_id=:rv, updated_at=now() WHERE
id=:id"),
                {"rv": version_id, "id": app["id"]},
            )
            db.commit()

    cover_letter = None
    if settings_row.get("ai_optimize_cover_letter") and jd:
        cover_letter = generate_cover_letter(db, app["user_id"], resume_text, jd)
        if cover_letter:
            attach_cover_letter(db, version_id, cover_letter)

    local_path = _download_resume(payload)

    account = _load_account(db, app)

    return apply_base.ApplyContext(
        user_id=app["user_id"],
        application_id=app["id"],
        posting=app,
        payload=payload,
        resume_text=resume_text,
        cover_letter=cover_letter,
        resume_local_path=local_path,
```

```
        settings=settings_row,
        account=account,
    )


def _load_account(db: Session, app: dict) -> dict | None:
    """解密平台帳號（按投遞來源匹配，通用規則）。

  規則：先按 platform=來源 取第一個可用帳號，再回退第一個 official 帳號。
  返回 {id, login, password}。
    """
    from app.core.security import decrypt_field, decrypt_json, get_dek_for_user

    dek = None

    def _decrypt(row) -> dict | None:
        nonlocal dek
        try:
            if dek is None:
                dek = get_dek_for_user(db, app["user_id"])
            return {
                "id": str(row["id"]),
                "login": decrypt_field(dek, row["login_identifier_enc"]),
                "password": decrypt_field(dek, row["password_enc"]) or "",
            }
        except Exception as exc:  # noqa: BLE001
            logger.warning("平台帳號解密失敗 user=%s: %s", app["user_id"], exc)
            return None

    # 以平台名文本比對（platform::text），避免任意來源值造成枚舉轉換失敗而中止事務
    for platform in (app["source"], "official"):
        if not platform:
            continue
        row = db.execute(
            text(
                "SELECT id, login_identifier_enc, password_enc FROM platform_accounts "
                "WHERE user_id=:uid AND platform::text = :p AND status != 'locked' "
                "ORDER BY created_at LIMIT 1"
            ),
            {"uid": app["user_id"], "p": platform},
        ).mappings().first()
        if row:
            return _decrypt(row)
    return None


def _download_resume(payload: dict) -> str | None:
    rf = payload.get("resume_file")
    if not rf:
        return None
```

```
    data = download_bytes(rf["storage_path"])
    ext = os.path.splitext(rf["name"])[1] or ".pdf"
    fd, path = tempfile.mkstemp(suffix=ext, prefix="jobfly_resume_")
    with os.fdopen(fd, "wb") as f:
        f.write(data)
    return path


# ── 4. 主流程
─────────────────────────────────────────────────────────────────────────────
──

def run_application(db: Session, application_id: str, dry_run: bool = False, force: bool =
False) -> dict:
    app = _load_application(db, application_id)
    if not app:
        return {"error": "application not found"}
    if app["status"] in ("submitted", "aborted", "not_supported", "duplicate_skip"):
        return {"skipped": app["status"]}
    if app["status"] == "in_progress":
        return {"skipped": "in_progress"}
    if app["status"] == "needs_input" and not force:
        return {"skipped": "needs_input"}

    attempt_id = db.execute(
        text(
            "INSERT INTO application_attempts (user_id, application_id, attempt_no, result) "
            "VALUES (:uid, :aid, :n, 'failed') RETURNING id"
        ),
        {"uid": app["user_id"], "aid": application_id, "n": app["attempts_count"] + 1},
    ).mappings().one()["id"]
    db.execute(
        text(
            "UPDATE applications SET status='in_progress', current_step='prepare', "
            "attempts_count=attempts_count+1, updated_at=now() WHERE id=:id"
        ),
        {"id": application_id},
    )
    db.commit()

    outcome: dict = {}
    local_path: str | None = None
    try:
        if not app.get("apply_url"):
            _finish(db, app, application_id, "not_supported", "該崗位無申請入口", attempt_id,
"aborted")
            outcome = {"status": "not_supported"}
        else:
            payload = build_candidate_payload(db, app["user_id"])
            _merge_answers(db, application_id, payload)
            blocker = _preflight(payload)
            if blocker:
                outcome = _block(db, app, application_id, attempt_id, blocker)
            else:
```